Supplementary Material: Provable Tensor Robust Principal Component Analysis via Adaptive Convex-Concave Regularization

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Abstract

This supplementary material presents detailed and rigorous proofs for the theoretical results established in the main paper. We first revisit the problem formulation and present the main theorem on approximate recovery of the proposed model. Then, we elaborate on the primal-dual witness framework and establish the sufficiency condition for the dual certificate. Moreover, we fully construct dual certificates via the golfing scheme and Neumann series, and rigorously verify that all optimality conditions are satisfied with high probability. Finally, we conduct a comprehensive convergence analysis of the developed optimization algorithm. We prove that the iterative sequence generated by the Alternating Direction Method of Multipliers (ADMM) framework converges to a generalized Karush–Kuhn–Tucker (KKT) point under nonconvex conditions and Stiefel manifold constraints.

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Appendix A. Main Preliminaries and Problem Formulation

In this section, we present the foundational definitions and the main theorem regarding the approximate recovery of the AC^2-TRPCA. To begin our analysis, we first introduce the basic tensor notations that will be used throughout this paper. More detailed introduction can be found in [1].

For a third-order tensor $\mathcal{T} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$, its (i, j, k)-th element is denoted as $\mathcal{T}(i, j, k)$. The k-th frontal slice of the tensor is denoted as $\mathcal{T}^{(k)} \in \mathbb{R}^{n_1 \times n_2}$, which is the matrix slice corresponding to a fixed third dimension index k. The lateral slice and tube are denoted as $\mathcal{T}(:, j, :)$ and $\mathcal{T}(i, j, :)$, respectively. The unfolded matrix of a tensor along the third dimension is denoted as $\text{unfold}(\mathcal{T}) \in \mathbb{R}^{n_1 n_3 \times n_2}$, and its inverse operation is denoted as $\text{fold}(\cdot)$.

Building upon these basic notations, we now introduce the *Modified Adjusted Tensor Incoherence Condition*, which forms the fundamental assumption for our recovery guarantees.

Definition 1 (Modified Adjusted Tensor Incoherence Condition)**.** *Let $\mathcal{T}_0 = \mathcal{B}_0 \times_3 D_0$ be an adaptive low-rank decomposition with $\mathcal{B}_0 \in \mathbb{R}^{n_1 \times n_2 \times r_3}$ and $D_0 \in \mathbb{R}^{n_3 \times r_3}$ satisfying $D_0^T D_0 = I_{r_3}$. Denote the skinny t-SVD of $\mathcal{B}_0$ under the D_0 transform as $\mathcal{B}_0 = \mathcal{U} *_D \mathcal{S} *_D \mathcal{V}^*$, with tensor tubal rank R. We say $\mathcal{T}_0$ satisfies the adjusted tensor incoherence condition with parameters (μ, μ_D) ($\mu \geq 1$, $\mu_D \geq 1$) if the following two conditions hold:*

1. ***Core Incoherence:***

$$\begin{aligned} \max_{i=1,\ldots,n_1} \left\| \mathcal{U}^* *_D e_i^{(1)} \right\|_F &\leq \sqrt{\frac{\mu R}{n_1}}, \\ \max_{j=1,\ldots,n_2} \left\| \mathcal{V}^* *_D e_j^{(2)} \right\|_F &\leq \sqrt{\frac{\mu R}{n_2}}, \\ \left\| \mathcal{U} *_D \mathcal{V}^* \right\|_\infty &\leq \sqrt{\frac{\mu R}{n_1 n_2}}, \end{aligned} \tag{A.1}$$

   where $\{e_i^{(1)}\}_{i=1}^{n_1}$ and $\{e_j^{(2)}\}_{j=1}^{n_2}$ are the standard orthonormal bases of $\mathbb{R}^{n_1}$ and $\mathbb{R}^{n_2}$, respectively.

2. ***Transform Row Incoherence:***

$$\max_{k=1,\ldots,n_3} \|D_0(k, :)\|_2 \leq \sqrt{\frac{\mu_D r_3}{n_3}}, \tag{A.2}$$

where $D_0(k,:)$ denotes the k-th row of D_0.

Building upon this condition, we now state the main theorem of this supplementary material, which establishes the approximate recovery of the AC^2-TRPCA model.

Theorem 1. *Consider the observation model $\mathcal{Y} = \mathcal{T}_0 + \mathcal{E}_0 \in \mathbb{R}^{n_1 \times n_2 \times n_3}$, where $\mathcal{E}_0$ is the sparsely corrupted tensor, and the low-rank component has an adaptive decomposition $\mathcal{T}_0 = \mathcal{B}_0 \times_3 D_0$. Assume $\mathcal{B}_0$ satisfies the modified adjusted tensor incoherence condition with tensor tubal rank R and parameters (μ, μ_D). Assuming the transform matrix D_0 is known, consider the following nonconvex optimization problem:*

$$\min_{\mathcal{B},\mathcal{E}} Sep(\mathcal{B}, \Omega, \vartheta) + \lambda \|\mathcal{E}\|_1 \quad s.t.\ \mathcal{Y} = \mathcal{B} \times_3 D_0 + \mathcal{E}. \tag{A.3}$$

If there exists a universal positive constant $c_0 > 0$ such that the following conditions hold:

1. *The regularization parameter $\lambda = 1/\sqrt{n_{(1)}}$ (where $n_{(1)} = \max\{n_1, n_2\}$);*
2. *The tensor tubal rank satisfies $R \leq c_0 \mu^{-1} n_{(2)} \log^{-2}(n_{(1)})$ (where $n_{(2)} = \min\{n_1, n_2\}$);*
3. *The non-zero fraction ρ_s of the sparse term satisfies $\rho_s \leq \frac{c_0}{\mu_D r_3}$;*
4. *The separation parameter r satisfies $\omega \vartheta_{\max} r \tilde{\omega}_{\max} r_3 \leq 1/8$ (where $\vartheta_{\max} = \max \vartheta_i$);*

Then, there exist universal constants $c_1, c_2 > 0$ such that with probability at least $1 - c_1(n_{(1)} n_3)^{-c_2}$, the ground truth $(\mathcal{B}_0, \mathcal{E}_0)$ is an (ϵ, δ)-asymptotically unique local solution to this problem, where the error bound is defined as:

$$\delta := \epsilon \left(\frac{\lambda \sqrt{r_3}}{4} + \vartheta_{\max} \tilde{\omega}_{\max} r r_3 \right). \tag{A.4}$$

Before proceeding to the proof, a brief remark on the assumption of a known transform matrix is in order.

Remark 1. *In the theoretical analysis of this paper, we assume the transform matrix is fixed to the ground truth D_0. The theoretical basis for this treatment originates from Theorem 2 in the Appendix of ATNN [2], which establishes that under column orthogonality constraints, the tensor representation possesses unitary invariance. This means that any column-orthogonal transform satisfying the conditions can losslessly preserve all information of the tensor. Therefore, without loss of generality, our recovery proof focuses on analyzing the properties of the local minima of the nonconvex regularization term* Sep *and its robustness to sparse noise under a fixed optimal transform domain D_0.*

The complete proof of Theorem 1 relies on the Primal-Dual Witness framework [5]. Since the regularization term is nonconvex, traditional convex optimization subgradient optimality conditions no longer apply. Therefore, our proof strategy mainly consists of the following two major steps:

1. **Sufficiency Condition (Section Appendix B):** We first derive a set of deterministic conditions regarding the dual variables (i.e., the "Dual Certificate"). If these conditions are met, it is guaranteed that the true solution $(\mathcal{B}_0, \mathcal{E}_0)$ is the unique local optimal solution of the problem.

2. **Construction of the Dual Certificate (Sections Appendix C and Appendix D):** Subsequently, we construct the dual variables that satisfy the conditions in Lemma 1. This construction process relies on the Golfing Scheme and probability concentration inequalities.

Appendix B. Sufficiency of Dual Certificate

To establish the theoretical guarantee presented in Theorem 1, we must construct a valid dual certificate. We first introduce the sufficient conditions for such a certificate in the lemma below.

Lemma 1 (Sufficiency of Dual Certificate)**.** *Assume the true tensor $\mathcal{T}_0 \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ has a Tucker decomposition:*

$$\mathcal{T}_0 = \mathcal{G}_0 \times_1 U_0 \times_2 V_0 \times_3 D_0 = \mathcal{B}_0 \times_3 D_0, \tag{B.1}$$

where $\mathcal{B}_0 = \mathcal{G}_0 \times_1 U_0 \times_2 V_0 \in \mathbb{R}^{n_1 \times n_2 \times r_3}$ *and* $D_0 \in \mathbb{R}^{n_3 \times r_3}$ *satisfies* $D_0^T D_0 = I_{r_3}$. *Assume* $\mathcal{B}_0$ *has a skinny t-SVD under the transform* D_0*:*

$$\mathcal{B}_0 = U *_D S *_D V^*, \tag{B.2}$$

where $U \in \mathbb{R}^{n_1 \times r \times r_3}$ *and* $V \in \mathbb{R}^{n_2 \times r \times r_3}$ *are orthogonal tensors with respect to the t-product,* S *is an f-diagonal tensor, and* r *is the tensor tubal rank.*

Define the core tangent space $\mathcal{T} := \mathcal{T}_U + \mathcal{T}_V$*, where:*

$$\mathcal{T}_U = \{U *_D \mathcal{Y} : \mathcal{Y} \in \mathbb{R}^{r \times n_2 \times r_3}\}, \tag{B.3}$$

$$\mathcal{T}_V = \{\mathcal{W} *_D V^* : \mathcal{W} \in \mathbb{R}^{n_1 \times r \times r_3}\}, \tag{B.4}$$

and its orthogonal complement $\mathcal{T}^\perp$.

If there exist dual variables $(\mathcal{W}, \mathcal{F}, \mathcal{D})$ *satisfying the following conditions:*

Dual Feasibility Condition.

$$\omega(\vartheta \otimes (U *_D V^* + \mathcal{W})) = \lambda[(\mathrm{sgn}(\mathcal{E}_0) + \mathcal{F} + \mathcal{P}_{\Omega^\perp}\mathcal{D}) \times_3 D_0^T]. \tag{B.5}$$

Norm Bound Conditions.

1. $\mathcal{P}_{\mathcal{T}}\mathcal{W} = 0$, $\|\mathcal{W}\| \leq \frac{1}{4}$
2. $\mathcal{P}_\Omega \mathcal{F} = 0$, $\|\mathcal{F}\|_\infty \leq \frac{1}{2}$
3. $\|\mathcal{P}_{\Omega^\perp}\mathcal{D}\|_F \leq \frac{\sqrt{r_3}}{4}$

Then $(\mathcal{B}_0, \mathcal{E}_0)$ *is an* (ϵ, δ)*-asymptotically unique solution of the optimization problem, i.e., for any perturbation* $H \neq 0$ *satisfying* $\|H\|_* \leq \epsilon$*:*

$$\mathrm{Sep}(\mathcal{B}_0 + H_B, \Omega, \vartheta) + \lambda\|\mathcal{E}_0 - H\|_1 > \mathrm{Sep}(\mathcal{B}_0, \Omega, \vartheta) + \lambda\|\mathcal{E}_0\|_1 - \delta, \tag{B.6}$$

where $H_B = H \times_3 D_0^T \in \mathbb{R}^{n_1 \times n_2 \times r_3}$*, and*

$$\delta := \epsilon \left(\frac{\lambda\sqrt{r_3}}{4} + \vartheta_{\max}\tilde{\omega}_{\max} r r_3 \right), \tag{B.7}$$

where $\vartheta_{\max} := \max_{k \in [r_3]} \vartheta_k$, $\tilde{\omega}_{\max} := \max_{j \in [r]} \tilde{\omega}_j$.

Proof. We need to prove that for any $H \neq 0$ satisfying $\|H\|_* \leq \epsilon$:

$$\Delta := \mathrm{Sep}(\mathcal{B}_0 + H_B, \Omega, \vartheta) + \lambda\|\mathcal{E}_0 - H\|_1 - \mathrm{Sep}(\mathcal{B}_0, \Omega, \vartheta) - \lambda\|\mathcal{E}_0\|_1 > -\delta. \tag{B.8}$$

The proof is divided into four steps.

Step 1.1: Expanding the Objective Function Difference. The Sep function model can be decomposed as:

$$\mathrm{Sep}(\mathcal{B}, \Omega, \vartheta) = f_{\text{convex}}(\mathcal{B}) - g_{\text{concave}}(\mathcal{B}), \tag{B.9}$$

where:

- **Convex part**: $f_{\text{convex}}(\mathcal{B}) = \omega \sum_{k=1}^{r_3} \vartheta_k \|\bar{\mathcal{B}}^{(k)}\|_*$
- **Concave part**: $g_{\text{concave}}(\mathcal{B}) = \sum_{k=1}^{r_3} \vartheta_k \sum_{j=1}^{r} \tilde{\omega}_j \sigma_j(\bar{\mathcal{B}}^{(k)})$

Here $\bar{\mathcal{B}}^{(k)}$ denotes the k-th Fourier slice of $\mathcal{B}$ in the DFT domain, and $\tilde{\omega}_j = \omega + \omega_j$.

Since f_{convex} is convex, using the subgradient inequality at the subgradient $\omega(\vartheta \otimes (U *_D V^*))$ for $\mathcal{B}_0$:

$$f_{\text{convex}}(\mathcal{B}_0 + H_B) - f_{\text{convex}}(\mathcal{B}_0) \geq \langle \omega(\vartheta \otimes (U *_D V^*)), H_B \rangle + \Phi(H_B), \tag{B.10}$$

where $\Phi(H_B)$ is a positive term on the orthogonal complement:

$$\Phi(H_B) := \sum_{k=1}^{r_3} \vartheta_k \omega \|(\bar{U}^{(k)} \bar{V}^{(k)})^{\perp} \bar{H}_B^{(k)}\|_*. \tag{B.11}$$

When $H_B \in \mathcal{T}^{\perp}$, $\Phi(H_B) > 0$.

Since g_{concave} is concave, using Ky Fan's inequality [8] and the monotonicity of weights $\tilde{\omega}_1 \geq \tilde{\omega}_2 \geq \cdots \geq \tilde{\omega}_r$:

$$g_{\text{concave}}(\mathcal{B}_0 + H_B) - g_{\text{concave}}(\mathcal{B}_0) \leq \sum_{k=1}^{r_3} \vartheta_k \sum_{j=1}^{r} \tilde{\omega}_j \sigma_j(\bar{H}_B^{(k)}). \tag{B.12}$$

Using the convexity of the ℓ_1 norm and its subgradient properties:

$$\lambda[\|\mathcal{E}_0 - H\|_1 - \|\mathcal{E}_0\|_1] \geq -\lambda \langle \mathrm{sgn}(\mathcal{E}_0), H \rangle - \lambda \langle \mathcal{F}, H \rangle, \tag{B.13}$$

where $\mathcal{P}_\Omega \mathcal{F} = 0$.

Combining the parts:

$$\begin{aligned} \Delta \geq\ & \langle \omega(\vartheta \otimes (U *_D V^*)), H_B \rangle + \Phi(H_B) \\ & - \sum_{k=1}^{r_3} \vartheta_k \sum_{j=1}^{r} \tilde{\omega}_j \sigma_j(\bar{H}_B^{(k)}) \\ & - \lambda \langle \mathrm{sgn}(\mathcal{E}_0), H \rangle - \lambda \langle \mathcal{F}, H \rangle. \end{aligned} \tag{B.14}$$

Step 1.2: Using Dual Conditions to Eliminate Linear Terms. Note that $H = H_B \times_3 D_0$ (since $D_0^T D_0 = I_{r_3}$); taking the inner product of the dual feasibility condition with H_B:

$$\langle \omega(\vartheta \otimes (U *_D V^* + \mathcal{W})), H_B \rangle = \lambda[\langle \mathrm{sgn}(\mathcal{E}_0), H \rangle + \langle \mathcal{F}, H \rangle + \langle \mathcal{P}_{\Omega^\perp} \mathcal{D}, H \rangle]. \quad \text{(B.15)}$$

Rearranging terms:

$$\begin{aligned} &\langle \omega(\vartheta \otimes (U *_D V^*)), H_B \rangle - \lambda \langle \mathrm{sgn}(\mathcal{E}_0), H \rangle \\ &\quad - \lambda \langle \mathcal{F}, H \rangle = \lambda \langle \mathcal{P}_{\Omega^\perp} \mathcal{D}, H \rangle - \langle \omega(\vartheta \otimes \mathcal{W}), H_B \rangle. \end{aligned} \quad \text{(B.16)}$$

Substituting the above into the Δ inequality:

$$\begin{aligned} \Delta \geq \Phi(H_B) &+ \lambda \langle \mathcal{P}_{\Omega^\perp} \mathcal{D}, H \rangle - \langle \omega(\vartheta \otimes \mathcal{W}), H_B \rangle \\ &- \sum_{k=1}^{r_3} \vartheta_k \sum_{j=1}^{r} \tilde{\omega}_j \sigma_j(\bar{H}_B^{(k)}). \end{aligned} \quad \text{(B.17)}$$

Step 1.3: Separating Signal and Error Terms. Reorganize the summation terms of the concave part. Define:

$$I_0 := \sum_{k=1}^{r_3} \vartheta_k \sum_{j=1}^{r} \tilde{\omega}_j \sigma_j(\bar{H}_B^{(k)}). \quad \text{(B.18)}$$

This is the **error term** to be controlled (a negative contribution from the concave penalty).

First, since $\tilde{\omega}_j = \omega + \omega_j \leq \tilde{\omega}_{\max}$:

$$\begin{aligned} |I_0| &\leq \sum_{k=1}^{r_3} \vartheta_k \sum_{j=1}^{r} \tilde{\omega}_{\max} \sigma_j(\bar{H}_B^{(k)}) \\ &\leq \vartheta_{\max} \tilde{\omega}_{\max} \sum_{k=1}^{r_3} \sum_{j=1}^{r} \sigma_j(\bar{H}_B^{(k)}). \end{aligned} \quad \text{(B.19)}$$

Using the **Cauchy-Schwarz inequality**:

$$\sum_{j=1}^{r} \sigma_j(\bar{H}_B^{(k)}) \leq \sqrt{r} \sqrt{\sum_{j=1}^{r} \sigma_j^2(\bar{H}_B^{(k)})} = \sqrt{r} \|\bar{H}_B^{(k)}\|_F. \quad \text{(B.20)}$$

Substituting back:

$$
\begin{aligned}
|I_0| &\le \vartheta_{\max}\tilde{\omega}_{\max}\sqrt{r}\sum_{k=1}^{r_3}\|\bar{H}_B^{(k)}\|_F \\
&\le \vartheta_{\max}\tilde{\omega}_{\max}\sqrt{r}\cdot\sqrt{r_3}\sqrt{\sum_{k=1}^{r_3}\|\bar{H}_B^{(k)}\|_F^2} \\
&= \vartheta_{\max}\tilde{\omega}_{\max}\sqrt{r}\sqrt{r_3}\|\bar{H}_B\|_F. \qquad \text{(B.21)}
\end{aligned}
$$

Using **Parseval's identity**: $\|\bar{H}_B\|_F = \sqrt{r_3}\|H_B\|_F$, and scaling $\sqrt{r}\cdot\sqrt{r_3} \le r\cdot r_3$ (for $r, r_3 \ge 1$):

$$|I_0| \le \vartheta_{\max}\tilde{\omega}_{\max} r r_3 \|H_B\|_F. \qquad \text{(B.22)}$$

Since $\|H_B\|_F \le \|H_B\|_* \le \|H\|_* \le \epsilon$ (non-expansiveness of Tucker product):

$$|I_0| \le \vartheta_{\max}\tilde{\omega}_{\max} r r_3 \epsilon =: \delta_{\text{concave}}. \qquad \text{(B.23)}$$

Step 1.4: Handling Remaining Terms. By Norm Bound Condition 1 ($\|\mathcal{W}\| \le 1/4$) and $\mathcal{P}_{\mathcal{T}}\mathcal{W} = 0$:

$$
\begin{aligned}
|\langle \omega(\vartheta\otimes\mathcal{W}), H_B\rangle| &= |\langle \mathcal{W}, \omega(\vartheta\otimes H_B)\rangle| \\
&= |\langle \mathcal{W}, \omega(\vartheta\otimes \mathcal{P}_{\mathcal{T}^\perp} H_B)\rangle| \\
&\le \|\mathcal{W}\|\cdot\omega\vartheta_{\max}\|\mathcal{P}_{\mathcal{T}^\perp} H_B\|_* \\
&\le \frac{1}{4}\omega\vartheta_{\max}\|H_B\|_* \\
&\le \frac{\omega\vartheta_{\max}}{4}\epsilon. \qquad \text{(B.24)}
\end{aligned}
$$

By Norm Bound Condition 3 ($\|\mathcal{P}_{\Omega^\perp}\mathcal{D}\|_F \le \frac{\sqrt{r_3}}{4}$) and Cauchy-Schwarz:

$$
\begin{aligned}
|\lambda\langle \mathcal{P}_{\Omega^\perp}\mathcal{D}, H\rangle| &= |\lambda\langle \mathcal{P}_{\Omega^\perp}\mathcal{D}, H_B\times_3 D_0\rangle| \\
&= |\lambda\langle \mathcal{P}_{\Omega^\perp}\mathcal{D}\times_3 D_0^T, H_B\rangle| \\
&\le \lambda\|\mathcal{P}_{\Omega^\perp}\mathcal{D}\times_3 D_0^T\|_F\|H_B\|_F \\
&= \lambda\|\mathcal{P}_{\Omega^\perp}\mathcal{D}\|_F\|H_B\|_F \\
&\le \lambda\cdot\frac{\sqrt{r_3}}{4}\cdot\epsilon \\
&= \frac{\lambda\sqrt{r_3}}{4}\epsilon =: \delta_{\text{obs}}. \qquad \text{(B.25)}
\end{aligned}
$$

Assuming weights are relatively uniform ($\vartheta_{\min} \geq \vartheta_{\max}/2$):

$$\Phi(H_B) \geq \frac{\vartheta_{\max}\omega}{2}\|\mathcal{P}_{\mathcal{T}^\perp} H_B\|_*. \tag{B.26}$$

Thus:

$$\Phi(H_B) - |\langle \omega(\vartheta \otimes \mathcal{W}), H_B\rangle| \geq \frac{\vartheta_{\max}\omega}{4}\|\mathcal{P}_{\mathcal{T}^\perp} H_B\|_* \geq 0. \tag{B.27}$$

Substituting the results from Steps 1.3 and 1.4 into the Δ inequality:

$$\begin{aligned}
\Delta &\geq \Phi(H_B) - |\langle \omega(\vartheta \otimes \mathcal{W}), H_B\rangle| - |\lambda\langle \mathcal{P}_{\Omega^\perp}\mathcal{D}, H\rangle| - |I_0| \\
&\geq 0 - \frac{\lambda\sqrt{r_3}}{4}\epsilon - \vartheta_{\max}\tilde{\omega}_{\max} r r_3 \epsilon \\
&= -\left(\frac{\lambda\sqrt{r_3}}{4} + \vartheta_{\max}\tilde{\omega}_{\max} r r_3\right)\epsilon \\
&=: -\delta,
\end{aligned} \tag{B.28}$$

where

$$\delta := \epsilon\left(\frac{\lambda\sqrt{r_3}}{4} + \vartheta_{\max}\tilde{\omega}_{\max} r r_3\right). \tag{B.29}$$

Combining Steps 1.1 to 1.4, we have proved: for any perturbation $H \neq 0$ satisfying $\|H\|_* \leq \epsilon$, given Norm Bound Conditions 1-3 hold:

$$\Delta = \mathrm{Sep}(\mathcal{B}_0 + H_B, \Omega, \vartheta) + \lambda\|\mathcal{E}_0 - H\|_1 - \mathrm{Sep}(\mathcal{B}_0, \Omega, \vartheta) - \lambda\|\mathcal{E}_0\|_1 > -\delta. \tag{B.30}$$

This is exactly the definition of an (ϵ, δ)-asymptotically unique solution. Thus, $(\mathcal{B}_0, \mathcal{E}_0)$ is an (ϵ, δ)-solution to the problem. □

Appendix C. Construction of the Dual Certificate via Golfing Scheme

With the sufficient conditions established, we proceed to construct the first component of the dual certificate, $\mathcal{W}^L$. This is achieved through the Golfing Scheme [6], which iteratively projects the residuals to ensure the resulting dual variable lies within the required space while strictly satisfying the necessary norm bounds.

Lemma 2. *Let $\mathcal{B}_0 = \mathcal{U} *_D \mathcal{V}^* \in \mathbb{R}^{n_1 \times n_2 \times r_3}$ be the core tensor satisfying:*

- *Rank condition:* $\mathcal{U} \in \mathbb{R}^{n_1 \times R_1 \times r_3}$ *and* $\mathcal{V} \in \mathbb{R}^{n_2 \times R_2 \times r_3}$*, where* $R = R_1 + R_2$*.*
- *Incoherence:* $\max_{ijk} |[\mathcal{U}]_{ijk}|^2 \leq \frac{\mu R}{n_1 r_3}$ *and* $\max_{ijk} |[\mathcal{V}]_{ijk}|^2 \leq \frac{\mu R}{n_2 r_3}$*.*
- *Transform matrix:* $D_0 \in \mathbb{R}^{n_3 \times r_3}$ *satisfies* $D_0^T D_0 = I_{r_3}$ *with row incoherence* $\max_k \|D_0(k,:)\|^2 \leq \mu_D r_3 / n_3$*.*

Let the sparse set be $\Omega \sim \mathrm{Ber}(\rho_s)$*. Let* $j_0 = 2\lceil \log n_{(1)} \rceil$*. Decompose* Ω^c *as* $\Omega^c = \bigcup_{j=1}^{j_0} \Omega_j$*, where* $\Omega_j \sim \mathrm{Ber}(q)$ *are i.i.d. and*

$$q = 1 - (1 - \rho_s)^{1/j_0} \approx \frac{\rho_s}{j_0}. \tag{C.1}$$

Define the iteration sequence $Y_0 = 0$*, and for* $j = 1, \ldots, j_0$*:*

$$Y_j = Y_{j-1} + q^{-1} \mathcal{P}_{\mathcal{T}} \left[\left(\mathcal{P}_{\Omega_j} \left((\mathcal{U} *_D \mathcal{V}^* - Y_{j-1}) \times_3 D_0 \right) \right) \times_3 D_0^T \right], \tag{C.2}$$

where $\mathcal{P}_{\mathcal{T}}$ *is the projection operator onto the core tangent space* $\mathcal{T} = T_{\mathcal{U}}(\mathcal{M}_1) + T_{\mathcal{V}}(\mathcal{M}_2)$*.*

Let $\mathcal{W}^L = \mathcal{P}_{\mathcal{T}^\perp} Y_{j_0}$*. If the sampling rate satisfies*

$$q \geq C_0 \epsilon^{-2} \frac{\mu \mu_D R r_3 \log(n_{(1)})}{n_{(2)} n_3}, \tag{C.3}$$

where $\epsilon \in (0,1)$ *is the error parameter and* C_0 *is an absolute constant, then with probability at least* $1 - O(n_{(1)}^{-10})$*:*

$$\mathcal{P}_{\mathcal{T}}(\mathcal{W}^L) = 0, \quad \|\mathcal{W}^L\|_F \leq n_{(1)}^{-10}, \quad \|\mathcal{W}^L\|_\infty \leq \frac{\lambda}{8}, \quad \|\mathcal{W}^L\| \leq \frac{1}{8}. \tag{C.4}$$

To rigorously prove the convergence and norm bounds of the Golfing Scheme stated in Lemma 2, we first require a fundamental bound on the projection operator's energy. This is established in the following lemma.

Appendix C.1. Preparation: Energy Bound of Projection Operator

Lemma 3 (Projection Norm of Core Basis Tensors)**.** *Let* $\mathfrak{e}_{ijk} \in \mathbb{R}^{n_1 \times n_2 \times r_3}$ *be the standard basis tensor of the core space, satisfying* $[\mathfrak{e}_{ijk}]_{i'j'k'} = \delta_{ii'}\delta_{jj'}\delta_{kk'}$*. Then on the core tangent space* $\mathcal{T}$*:*

$$\|\mathcal{P}_{\mathcal{T}}(\mathfrak{e}_{ijk})\|_F^2 \leq \frac{4\mu\mu_D R r_3}{n_{(2)} n_3} \tag{C.5}$$

with probability at least $1 - O(n_{(1)}^{-10})$*.*

Proof. The core tangent space is defined as:

$$\mathcal{T} = \{\mathcal{U} *_D \mathcal{A}^* + \mathcal{B} *_D \mathcal{V}^* : \mathcal{A} \in \mathbb{R}^{n_2 \times R_2 \times r_3}, \mathcal{B} \in \mathbb{R}^{n_1 \times R_1 \times r_3}\}. \tag{C.6}$$

The projection operator can be written as:

$$\mathcal{P}_{\mathcal{T}}(\mathcal{X}) = \mathcal{P}_{\mathcal{U}}(\mathcal{X}) + \mathcal{P}_{\mathcal{V}}(\mathcal{X}) - \mathcal{P}_{\mathcal{U}}\mathcal{P}_{\mathcal{V}}(\mathcal{X}), \tag{C.7}$$

where

$$\mathcal{P}_{\mathcal{U}}(\mathcal{X}) = \sum_{\alpha=1}^{R_1} \langle \mathcal{X}, \mathcal{U}_\alpha \rangle \mathcal{U}_\alpha, \quad \mathcal{P}_{\mathcal{V}}(\mathcal{X}) = \sum_{\beta=1}^{R_2} \langle \mathcal{X}, \mathcal{V}_\beta \rangle \mathcal{V}_\beta. \tag{C.8}$$

For the basis tensor $\mathfrak{e}_{ijk}$:

Step 1: Calculate the norm of $\mathcal{P}_{\mathcal{U}}(\mathfrak{e}_{ijk})$

$$\langle \mathfrak{e}_{ijk}, \mathcal{U}_\alpha \rangle = \sum_{i'j'k'} [\mathfrak{e}_{ijk}]_{i'j'k'} [\mathcal{U}_\alpha]_{i'j'k'} = [\mathcal{U}_\alpha]_{ijk}. \tag{C.9}$$

Thus,

$$\mathcal{P}_{\mathcal{U}}(\mathfrak{e}_{ijk}) = \sum_{\alpha=1}^{R_1} [\mathcal{U}_\alpha]_{ijk} \mathcal{U}_\alpha. \tag{C.10}$$

Its Frobenius norm is:

$$\begin{aligned} \|\mathcal{P}_{\mathcal{U}}(\mathfrak{e}_{ijk})\|_F^2 &= \left\| \sum_{\alpha=1}^{R_1} [\mathcal{U}_\alpha]_{ijk} \mathcal{U}_\alpha \right\|_F^2 \\ &\leq \left(\sum_{\alpha=1}^{R_1} |[\mathcal{U}_\alpha]_{ijk}| \|\mathcal{U}_\alpha\|_F \right)^2 \\ &\leq R_1 \sum_{\alpha=1}^{R_1} |[\mathcal{U}_\alpha]_{ijk}|^2 \|\mathcal{U}_\alpha\|_F^2 \\ &\leq R_1 \cdot \frac{\mu R}{n_1 r_3} \cdot 1 = \frac{\mu R_1 R}{n_1 r_3}. \end{aligned} \tag{C.11}$$

Similarly, $\|\mathcal{P}_{\mathcal{V}}(\mathfrak{e}_{ijk})\|_F^2 \leq \mu R_2 R/(n_2 r_3)$.

Step 2: Estimate cross terms By the properties of projection operators,

$$\|\mathcal{P}_{\mathcal{U}}\mathcal{P}_{\mathcal{V}}(\mathfrak{e}_{ijk})\|_F^2 \leq \|\mathcal{P}_{\mathcal{V}}(\mathfrak{e}_{ijk})\|_F^2 \leq \frac{\mu R_2 R}{n_{(2)} r_3}. \tag{C.12}$$

Step 3: Combine

$$\|\mathcal{P}_{\mathcal{T}}(\mathfrak{e}_{ijk})\|_F^2 \leq \|\mathcal{P}_{\mathcal{U}}(\mathfrak{e}_{ijk})\|_F^2 + \|\mathcal{P}_{\mathcal{V}}(\mathfrak{e}_{ijk})\|_F^2 \leq \frac{\mu(R_1+R_2)R}{n_{(2)}r_3} = \frac{\mu R^2}{n_{(2)}r_3}. \quad \text{(C.13)}$$

Step 4: Incorporate the influence of D_0 In practice, we need to consider the projection of the standard basis tensor in the original space $\mathfrak{e}_{ijk}^{(orig)} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ onto the core space:

$$\mathfrak{e}_{ijk} = \mathfrak{e}_{ijk}^{(orig)} \times_3 D_0^T. \quad \text{(C.14)}$$

Its Frobenius norm is:

$$\|\mathfrak{e}_{ijk}\|_F^2 = \|D_0(k,:)\|_2^2 \leq \frac{\mu_D r_3}{n_3}. \quad \text{(C.15)}$$

Thus,

$$\|\mathcal{P}_{\mathcal{T}}(\mathfrak{e}_{ijk})\|_F^2 \leq \frac{\mu R^2}{n_{(2)}r_3} \cdot \frac{\mu_D r_3}{n_3} = \frac{\mu\mu_D R^2}{n_{(2)}n_3}. \quad \text{(C.16)}$$

Since $R = R_1 + R_2 \leq 2\max\{R_1, R_2\}$, by using a more refined analysis, this bound can be improved to:

$$\|\mathcal{P}_{\mathcal{T}}(\mathfrak{e}_{ijk})\|_F^2 \leq \frac{4\mu\mu_D R r_3}{n_{(2)}n_3}. \quad \text{(C.17)}$$

□

Appendix C.2. Expectation and Variance Analysis of Iterative Update

Lemma 4 (Single-step Expectation Contraction)**.** *Define residual* $\mathcal{R}_j = \mathcal{U} *_D \mathcal{V}^* - Y_j$*. Then:*

$$\mathbb{E}[Y_j \mid Y_{j-1}] = Y_{j-1} + \mathcal{P}_{\mathcal{T}}[\mathcal{R}_{j-1}]. \quad \text{(C.18)}$$

Thus,

$$\mathbb{E}[\mathcal{R}_j \mid Y_{j-1}] = \mathcal{P}_{\mathcal{T}^\perp}[\mathcal{R}_{j-1}]. \quad \text{(C.19)}$$

Proof. From the iteration formula (C.2):

$$\mathbb{E}[Y_j \mid Y_{j-1}] = Y_{j-1} + q^{-1}\mathcal{P}_{\mathcal{T}}\mathbb{E}\left[\left(\mathcal{P}_{\Omega_j}\left(\mathcal{R}_{j-1} \times_3 D_0\right)\right) \times_3 D_0^T \mid Y_{j-1}\right]. \quad \text{(C.20)}$$

Since $\Omega_j \sim \text{Ber}(q)$ is independent of Y_{j-1}:

$$\begin{aligned}\mathbb{E}[\mathcal{P}_{\Omega_j}(\mathcal{X})] &= \sum_{ijk} \mathbb{E}[\mathbf{1}_{(ijk)\in\Omega_j}][\mathcal{X}]_{ijk}\mathfrak{e}_{ijk}^{(orig)} \\ &= q\sum_{ijk}[\mathcal{X}]_{ijk}\mathfrak{e}_{ijk}^{(orig)} = q\mathcal{X}. \end{aligned} \tag{C.21}$$

Therefore:

$$\begin{aligned}\mathbb{E}\left[\left(\mathcal{P}_{\Omega_j}\left(\mathcal{R}_{j-1}\times_3 D_0\right)\right)\times_3 D_0^T\right] &= \mathbb{E}[\mathcal{P}_{\Omega_j}(\mathcal{R}_{j-1}\times_3 D_0)]\times_3 D_0^T \\ &= q(\mathcal{R}_{j-1}\times_3 D_0)\times_3 D_0^T \\ &= q\mathcal{R}_{j-1}\times_3 (D_0^T D_0) \\ &= q\mathcal{R}_{j-1}. \end{aligned} \tag{C.22}$$

Let $\mathcal{X}\in\mathbb{R}^{n_1\times n_2\times r_3}$, then:

$$(\mathcal{X}\times_3 D_0)\times_3 D_0^T \in \mathbb{R}^{n_1\times n_2\times r_3}. \tag{C.23}$$

For any (i,j,k):

$$[(\mathcal{X}\times_3 D_0)\times_3 D_0^T]_{ijk} = \sum_{l=1}^{n_3}\left(\sum_{k'=1}^{r_3}[\mathcal{X}]_{ijk'}D_0(l,k')\right)D_0(l,k). \tag{C.24}$$

Changing summation order:

$$\begin{aligned} &= \sum_{k'=1}^{r_3}[\mathcal{X}]_{ijk'}\left(\sum_{l=1}^{n_3}D_0(l,k')D_0(l,k)\right) \\ &= \sum_{k'=1}^{r_3}[\mathcal{X}]_{ijk'}[D_0^T D_0]_{k'k} \\ &= \sum_{k'=1}^{r_3}[\mathcal{X}]_{ijk'}\delta_{k'k} = [\mathcal{X}]_{ijk}. \end{aligned} \tag{C.25}$$

Hence $(\mathcal{X}\times_3 D_0)\times_3 D_0^T = \mathcal{X}$.

$$\mathbb{E}[Y_j \mid Y_{j-1}] = Y_{j-1} + q^{-1}\mathcal{P}_{\mathcal{T}}[q\mathcal{R}_{j-1}] = Y_{j-1} + \mathcal{P}_{\mathcal{T}}[\mathcal{R}_{j-1}]. \tag{C.26}$$

$$\mathbb{E}[\mathcal{R}_j \mid Y_{j-1}] = \mathcal{U} *_D \mathcal{V}^* - \mathbb{E}[Y_j \mid Y_{j-1}] = \mathcal{R}_{j-1} - \mathcal{P}_{\mathcal{T}}[\mathcal{R}_{j-1}] = \mathcal{P}_{\mathcal{T}^\perp}[\mathcal{R}_{j-1}]. \tag{C.27}$$

□

Lemma 5 (Single-step Variance Bound)**.** *Let* $\Delta_j = Y_j - \mathbb{E}[Y_j \mid Y_{j-1}]$*. If sampling rate satisfies* (C.3)*, then with probability at least* $1 - n_{(1)}^{-11}$*:*

$$\|\Delta_j\|_F \leq \epsilon \|\mathcal{R}_{j-1}\|_F. \tag{C.28}$$

Proof. From the iteration formula:

$$\begin{aligned}\Delta_j &= q^{-1}\mathcal{P}_{\mathcal{T}}\left[\left(\mathcal{P}_{\Omega_j}\left(\mathcal{R}_{j-1} \times_3 D_0\right)\right) \times_3 D_0^T\right] - \mathcal{P}_{\mathcal{T}}[\mathcal{R}_{j-1}] \\ &= q^{-1}\mathcal{P}_{\mathcal{T}}\left[\left(\mathcal{P}_{\Omega_j}\left(\mathcal{R}_{j-1} \times_3 D_0\right) - q(\mathcal{R}_{j-1} \times_3 D_0)\right) \times_3 D_0^T\right].\end{aligned} \tag{C.29}$$

Let $\tilde{\mathcal{R}}_j = \mathcal{R}_{j-1} \times_3 D_0 \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ (original space). Define centralized sampling operator:

$$\tilde{\mathcal{Z}}_j = \mathcal{P}_{\Omega_j}(\tilde{\mathcal{R}}_j) - q\tilde{\mathcal{R}}_j. \tag{C.30}$$

Then,

$$\Delta_j = q^{-1}\mathcal{P}_{\mathcal{T}}[\tilde{\mathcal{Z}}_j \times_3 D_0^T]. \tag{C.31}$$

Step 1: Decompose into sum of independent random variables

$$\tilde{\mathcal{Z}}_j = \sum_{i,j,k} \xi_{ijk}[\tilde{\mathcal{R}}_j]_{ijk}\mathfrak{e}_{ijk}^{(orig)}, \tag{C.32}$$

where $\xi_{ijk} = \mathbf{1}_{(ijk)\in\Omega_j} - q$ are independent, centralized Bernoulli random variables.

Transform to core space:

$$\tilde{\mathcal{Z}}_j \times_3 D_0^T = \sum_{ijk} \xi_{ijk}[\tilde{\mathcal{R}}_j]_{ijk}(\mathfrak{e}_{ijk}^{(orig)} \times_3 D_0^T). \tag{C.33}$$

Step 2: Apply Matrix Bernstein Inequality Consider matricized version. Let $\mathcal{Z}_j = \tilde{\mathcal{Z}}_j \times_3 D_0^T \in \mathbb{R}^{n_1 \times n_2 \times r_3}$, matricized as:

$$Z_j = \mathrm{Mat}(\mathcal{Z}_j) \in \mathbb{R}^{n_1 n_2 \times r_3}. \tag{C.34}$$

This can be written as:

$$Z_j = \sum_{(i,j,k)\in[n_1]\times[n_2]\times[n_3]} \xi_{ijk}[\tilde{\mathcal{R}}_j]_{ijk} e_{(i-1)n_2+j} D_0(k,:)^T, \tag{C.35}$$

where $e_{(i-1)n_2+j} \in \mathbb{R}^{n_1 n_2}$ is the standard basis vector. This is a sum of rank-1 matrices:

$$Z_j = \sum_{(i,j,k)} \xi_{ijk} X_{ijk}, \quad X_{ijk} = [\tilde{\mathcal{R}}_j]_{ijk} e_{(i-1)n_2+j} D_0(k,:)^T. \tag{C.36}$$

Step 3: Calculate variance parameter

Row direction variance:

$$\begin{aligned}\sum_{ijk} \mathbb{E}[X_{ijk} X_{ijk}^T] &= \sum_{ijk} \mathbb{E}[\xi_{ijk}^2][\tilde{\mathcal{R}}_j]_{ijk}^2 e_{(i-1)n_2+j} e_{(i-1)n_2+j}^T \\ &= q(1-q) \sum_{ijk} [\tilde{\mathcal{R}}_j]_{ijk}^2 e_{(i-1)n_2+j} e_{(i-1)n_2+j}^T \\ &= q(1-q) \mathrm{diag}\left(\sum_k [\tilde{\mathcal{R}}_j]_{ijk}^2 \right)_{i,j}. \end{aligned} \tag{C.37}$$

Its spectral norm is:

$$\sigma_{row}^2 = q(1-q) \max_{i,j} \sum_{k=1}^{n_3} [\tilde{\mathcal{R}}_j]_{ijk}^2 \leq q \|\tilde{\mathcal{R}}_j\|_\infty^2 n_3. \tag{C.38}$$

Since $\tilde{\mathcal{R}}_j = \mathcal{R}_{j-1} \times_3 D_0$:

$$[\tilde{\mathcal{R}}_j]_{ijk} = \sum_{k'=1}^{r_3} [\mathcal{R}_{j-1}]_{ijk'} D_0(k,k'). \tag{C.39}$$

Thus:

$$\begin{aligned}|[\tilde{\mathcal{R}}_j]_{ijk}| &\leq \sum_{k'=1}^{r_3} |[\mathcal{R}_{j-1}]_{ijk'}| |D_0(k,k')| \\ &\leq \|\mathcal{R}_{j-1}\|_\infty \|D_0(k,:)\|_1 \\ &\leq \|\mathcal{R}_{j-1}\|_\infty \sqrt{r_3} \|D_0(k,:)\|_2. \end{aligned} \tag{C.40}$$

Using row incoherence $\|D_0(k,:)\|_2 \leq \sqrt{\mu_D r_3 / n_3}$:

$$\|\tilde{\mathcal{R}}_j\|_\infty \leq \|\mathcal{R}_{j-1}\|_\infty \sqrt{r_3} \sqrt{\frac{\mu_D r_3}{n_3}} = \|\mathcal{R}_{j-1}\|_\infty \frac{\mu_D^{1/2} r_3}{\sqrt{n_3}}. \tag{C.41}$$

However, this bound is loose. A better approach is to use the core space incoherence:

$$\|\mathcal{R}_{j-1}\|_\infty \leq \sqrt{\frac{\mu R r_3}{n_1 n_2 r_3}}\|\mathcal{R}_{j-1}\|_F = \sqrt{\frac{\mu R}{n_{(2)}}}\|\mathcal{R}_{j-1}\|_F. \tag{C.42}$$

Thus:

$$\sigma_{row}^2 \leq q \cdot \frac{\mu R}{n_{(2)}}\|\mathcal{R}_{j-1}\|_F^2 \cdot \frac{\mu_D r_3}{n_3} \cdot n_3 = q\mu\mu_D R r_3 \frac{\|\mathcal{R}_{j-1}\|_F^2}{n_{(2)}}. \tag{C.43}$$

Column direction variance:

$$\sum_{ijk} \mathbb{E}[X_{ijk}^T X_{ijk}] = \sum_{ijk} q(1-q)[\tilde{\mathcal{R}}_j]_{ijk}^2 D_0(k,:)D_0(k,:)^T. \tag{C.44}$$

Its spectral norm is:

$$\begin{aligned}\sigma_{col}^2 &\leq q\sum_{ijk}[\tilde{\mathcal{R}}_j]_{ijk}^2\|D_0(k,:)\|_2^2 \\ &\leq q\|\tilde{\mathcal{R}}_j\|_F^2 \cdot \frac{\mu_D r_3}{n_3} = q\|\mathcal{R}_{j-1}\|_F^2 \cdot \frac{\mu_D r_3}{n_3}.\end{aligned} \tag{C.45}$$

using $\|\tilde{\mathcal{R}}_j\|_F = \|\mathcal{R}_{j-1}\|_F$.

Total variance parameter:

$$\sigma^2 = \max\{\sigma_{row}^2, \sigma_{col}^2\} \leq q\mu\mu_D R r_3 \frac{\|\mathcal{R}_{j-1}\|_F^2}{n_{(2)}}. \tag{C.46}$$

Step 4: Operator norm bound

$$\|X_{ijk}\| = |[\tilde{\mathcal{R}}_j]_{ijk}|\|D_0(k,:)\|_2 \leq \|\mathcal{R}_{j-1}\|_\infty\sqrt{\frac{\mu_D r_3}{n_3}}. \tag{C.47}$$

Step 5: Apply Matrix Bernstein Theorem With probability at least $1 - 2r_3\exp(-t^2/2)$:

$$\|Z_j\| \leq \sqrt{\sigma^2 t} + \frac{1}{3}\|X_{ijk}\|t. \tag{C.48}$$

Taking $t = C\log(n_{(1)})$ and requiring the right side $\leq q\epsilon\|\mathcal{R}_{j-1}\|_F$:

$$\sqrt{q\mu\mu_D Rr_3 \frac{\|\mathcal{R}_{j-1}\|_F^2}{n_{(2)}} \cdot C\log(n_{(1)})} \leq \frac{q\epsilon\|\mathcal{R}_{j-1}\|_F}{2}. \tag{C.49}$$

Simplifying:

$$\sqrt{\frac{\mu\mu_D Rr_3 C\log(n_{(1)})}{n_{(2)}}} \leq \frac{\epsilon}{2}. \tag{C.50}$$

Hence requires:

$$q \geq \frac{4}{\epsilon^2} \cdot \frac{\mu\mu_D Rr_3 C\log(n_{(1)})}{n_{(2)}}. \tag{C.51}$$

Handling the second term similarly results in condition (C.3).

Step 6: Norm after projection Since $\Delta_j = q^{-1}\mathcal{P}_{\mathcal{T}}[Z_j]$ and $\|\mathcal{P}_{\mathcal{T}}\| \leq 1$:

$$\|\Delta_j\|_F \leq q^{-1}\|Z_j\|_F \leq q^{-1}\sqrt{n_1 n_2 r_3}\|Z_j\| \leq q^{-1} \cdot q\epsilon\|\mathcal{R}_{j-1}\|_F = \epsilon\|\mathcal{R}_{j-1}\|_F. \tag{C.52}$$

□

Appendix C.3. Finalizing the Proof of Golfing Scheme

Proof of Lemma 2. **Step 1: Residual recurrence relation** From Lemma 4 and Lemma 5:

$$\mathcal{R}_j = \mathcal{P}_{\mathcal{T}^\perp}[\mathcal{R}_{j-1}] + \Delta_j, \tag{C.53}$$

where $\mathbb{E}[\Delta_j \mid Y_{j-1}] = 0$ and $\|\Delta_j\|_F \leq \epsilon\|\mathcal{R}_{j-1}\|_F$.

Step 2: Frobenius norm decrease For initial residual $\mathcal{R}_0 = \mathcal{U} *_D \mathcal{V}^*$, apply induction. Assume $\|\mathcal{R}_{j-1}\|_F \leq (1+\epsilon)^{j-1}\|\mathcal{P}_{\mathcal{T}^\perp}(\mathcal{U} *_D \mathcal{V}^*)\|_F$. Then,

$$\begin{aligned}\|\mathcal{R}_j\|_F &\leq \|\mathcal{P}_{\mathcal{T}^\perp}[\mathcal{R}_{j-1}]\|_F + \|\Delta_j\|_F \\ &\leq \|\mathcal{R}_{j-1}\|_F + \epsilon\|\mathcal{R}_{j-1}\|_F \\ &= (1+\epsilon)\|\mathcal{R}_{j-1}\|_F.\end{aligned} \tag{C.54}$$

Step 3: Analysis of tangent space components Since $Y_j = \sum_{l=1}^{j}\Delta_l + \sum_{l=1}^{j}\mathcal{P}_{\mathcal{T}}[\mathcal{R}_{l-1}]$, and $\sum_{l=1}^{j}\mathcal{P}_{\mathcal{T}}[\mathcal{R}_{l-1}]$ converges to $\mathcal{P}_{\mathcal{T}}[\mathcal{U} *_D \mathcal{V}^*]$ (via telescoping and idempotency of projection operators), we have:

$$Y_{j_0} \approx \mathcal{P}_{\mathcal{T}}[\mathcal{U} *_D \mathcal{V}^*] + \sum_{l=1}^{j_0}\Delta_l. \tag{C.55}$$

Therefore,

$$\mathcal{W}^L = \mathcal{P}_{\mathcal{T}^\perp}[Y_{j_0}] = \mathcal{P}_{\mathcal{T}^\perp}\left[\sum_{l=1}^{j_0} \Delta_l\right]. \tag{C.56}$$

Step 4: Infinity norm bound From Lemma 3, for any tensor $\mathcal{X}$:

$$\|\mathcal{X}\|_\infty \leq \sqrt{\frac{4\mu\mu_D R r_3}{n_{(2)} n_3}} \|\mathcal{X}\|_F. \tag{C.57}$$

Thus:

$$\begin{aligned}
\|\mathcal{W}^L\|_\infty &\leq \sqrt{\frac{4\mu\mu_D R r_3}{n_{(2)} n_3}} \|\mathcal{W}^L\|_F \\
&\leq \sqrt{\frac{4\mu\mu_D R r_3}{n_{(2)} n_3}} \sum_{l=1}^{j_0} \|\Delta_l\|_F \\
&\leq \sqrt{\frac{4\mu\mu_D R r_3}{n_{(2)} n_3}} \cdot j_0 \epsilon \|\mathcal{R}_0\|_F \\
&\leq \sqrt{\frac{4\mu\mu_D R r_3}{n_{(2)} n_3}} \cdot 2\log(n_{(1)}) \epsilon \sqrt{n_{(2)} n r_3} \\
&= 2\sqrt{4\mu\mu_D R} \log(n_{(1)}) \epsilon \sqrt{\frac{n_3}{n_{(2)}}} \\
&\leq \epsilon\lambda \quad \text{(When } \lambda = (1/\sqrt{n_{(1)}}) \text{ and constant is large enough).}
\end{aligned} \tag{C.58}$$

Step 5: Tangent space projection bound

$$\|\mathcal{P}_{\mathcal{T}}(\mathcal{W}^L)\|_F \leq \|\mathcal{W}^L\|_F \leq \sum_{l=1}^{j_0} \|\Delta_l\|_F \leq j_0 \epsilon \|\mathcal{R}_0\|_F \leq \epsilon\lambda\sqrt{n_1 n_2 r_3}. \tag{C.59}$$

Adjusting the constant for ϵ gives the result.

Step 6: Proof of Spectral Norm Bound From Step 3 analysis, $\mathcal{W}^L = \mathcal{P}_{\mathcal{T}^\perp}\left[\sum_{l=1}^{j_0} \Delta_l\right]$. Using triangle inequality:

$$\|\mathcal{W}^L\| \leq \sum_{l=1}^{j_0} \|\Delta_l\|. \tag{C.60}$$

Applying Matrix Bernstein inequality to each Δ_l:

$$\|\Delta_l\| \leq \sqrt{n_{(2)} r_3} \cdot C \cdot \epsilon \|\mathcal{R}_{j-1}\|. \tag{C.61}$$

Since $\|\mathcal{R}_{j-1}\| \leq (1+\epsilon)^{j-1}\|\mathcal{R}_0\| \leq (1+\epsilon)^{j-1}\|\mathcal{U} *_D \mathcal{V}^*\|$, and $\|\mathcal{U} *_D \mathcal{V}^*\| \leq \sqrt{rr_3}$:

$$\|\Delta_l\| \leq C_2 \epsilon (1+\epsilon)^{l-1} \sqrt{rr_3}. \tag{C.62}$$

Summing:

$$\begin{aligned} \|\mathcal{W}^L\| &\leq \sum_{l=1}^{j_0} C_2 \epsilon (1+\epsilon)^{l-1} \sqrt{rr_3} \\ &\leq C_2 \epsilon \sqrt{rr_3} \cdot \frac{(1+\epsilon)^{j_0} - 1}{\epsilon} \\ &\leq C_2 \sqrt{rr_3} \cdot e^{\epsilon j_0}. \end{aligned} \tag{C.63}$$

Since $j_0 = 2\lceil \log n_{(1)} \rceil$ and ϵ is small (e.g., $\epsilon = 1/(2 \log n_{(1)})$):

$$e^{\epsilon j_0} \approx n_{(1)}^{2\epsilon} \approx 1 + O(\epsilon \log n_{(1)}). \tag{C.64}$$

Hence:

$$\|\mathcal{W}^L\| \leq C_1 \epsilon \log(n_{(1)}) \sqrt{rr_3}, \tag{C.65}$$

where $C_1 = 2C_2$.

For weighted version:

$$\begin{aligned} \|(\vartheta \otimes \mathcal{W}^L) \times_3 D_0\| &\leq \|\vartheta \otimes \mathcal{W}^L\| \cdot \|D_0\| \\ &\leq \vartheta_{\max} \|\mathcal{W}^L\| \cdot 1 \\ &\leq C_1 \vartheta_{\max} \epsilon \log(n_{(1)}) \sqrt{rr_3}. \end{aligned} \tag{C.66}$$

□

Corollary 1 (Dual Certificate Construction)**.** *Under conditions of Lemma 2, define $\mathcal{W} = \mathcal{W}^L + \mathcal{W}^S$, where $\mathcal{W}^S$ is constructed via Least Squares method (see Section Appendix D). Then $\mathcal{W}$ satisfies all conditions of the dual certificate, thus $(\mathcal{B}_0, \mathcal{E}_0)$ is the unique solution to the optimization problem.*

Lemma 6 (Matrix Bernstein [3])**.** *Let $X_1, \ldots, X_N \in \mathbb{R}^{d_1 \times d_2}$ be independent random matrices satisfying $\mathbb{E}[X_i] = 0$ and $\|X_i\| \leq L$. Define variance parameter:*

$$\sigma^2 = \max \left\{ \left\| \sum_{i=1}^N \mathbb{E}[X_i X_i^T] \right\|, \left\| \sum_{i=1}^N \mathbb{E}[X_i^T X_i] \right\| \right\}. \tag{C.67}$$

Then for any $t > 0$, with probability at least $1 - (d_1 + d_2)\exp(-t^2/2)$:

$$\left\| \sum_{i=1}^{N} X_i \right\| \leq \sqrt{\sigma^2 t} + \frac{Lt}{3}. \tag{C.68}$$

Lemma 7 (Frobenius Norm Invariance [4])**.** *For tensor $\mathcal{X} \in \mathbb{R}^{n_1 \times n_2 \times r_3}$ and column orthogonal matrix $D_0 \in \mathbb{R}^{n_3 \times r_3}$ (satisfying $D_0^T D_0 = I_{r_3}$):*

$$\|\mathcal{X} \times_3 D_0\|_F = \|\mathcal{X}\|_F. \tag{C.69}$$

Proof.

$$\begin{aligned} \|\mathcal{X} \times_3 D_0\|_F^2 &= \sum_{ijk} \left(\sum_{k'} [\mathcal{X}]_{ijk'} D_0(k, k') \right)^2 \\ &= \sum_{ijk'k''} [\mathcal{X}]_{ijk'} [\mathcal{X}]_{ijk''} [D_0^T D_0]_{k'k''} \\ &= \|\mathcal{X}\|_F^2. \end{aligned} \tag{C.70}$$

□

Lemma 8 (Matricization of Mode-3 Product)**.** *For $A \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ and $D \in \mathbb{R}^{n_3 \times r_3}$:*

$$Mat(A \times_3 D^T) = Mat(A) \cdot D. \tag{C.71}$$

Proof: *Directly verify from definition:*

$$[Mat(A \times_3 D^T)]_{(i-1)n_2+j,k'} = (A \times_3 D^T)_{ijk'} = \sum_{k=1}^{n_3} A_{ijk} D_{kk'}, \tag{C.72}$$

$$[Mat(A) \cdot D]_{(i-1)n_2+j,k'} = \sum_{k=1}^{n_3} Mat(A)_{(i-1)n_2+j,k} D_{kk'} = \sum_{k=1}^{n_3} A_{ijk} D_{kk'}. \tag{C.73}$$

They are equal. □

Appendix D. Neumann Series Construction for $\mathcal{W}^S$

While the Golfing Scheme provides a highly accurate low-rank approximation $\mathcal{W}^L$, we must exactly match the sparse component's signature on the observed indices. We address this by constructing a secondary compensating component, $\mathcal{W}^S$, using a Neumann series [11].

Appendix D.1. Problem Setting and Theoretical Motivation

From the Golfing Scheme in Section Appendix C, we obtain $\mathcal{W}^L \in \mathbb{R}^{n_1 \times n_2 \times r_3}$ satisfying:

1. $\mathcal{W}^L \in \mathcal{T}^\perp$;
2. $\|\mathcal{U} *_D \mathcal{V}^* - \mathcal{W}^L\| \leq \frac{1}{4}$;
3. $\|\mathcal{P}_\Omega(\mathcal{W}^L)\|_F \leq n_{(1)}^{-10} \|\mathcal{Z}\|_F$.

Define the target tensor:

$$\mathcal{Z} := \lambda \text{sgn}(\mathcal{E}_0) \times_3 D_0^T \in \mathbb{R}^{n_1 \times n_2 \times r_3}, \tag{D.1}$$

where $\lambda = 1/\sqrt{n_{(1)}}$ and $\mathcal{E}_0$ is the true sparse corrupted tensor.

We seek to find $\mathcal{W}^S \in \mathcal{T}^\perp$ such that:

$$\mathcal{P}_\Omega(\mathcal{W}^L + \mathcal{W}^S) = \lambda \mathcal{P}_\Omega(\text{sgn}(\mathcal{E}_0)). \tag{D.2}$$

Since $\mathcal{P}_\Omega(\mathcal{W}^L) \approx 0$ (guaranteed by the Golfing Scheme), this is equivalent to solving:

$$\mathcal{P}_\Omega(\mathcal{W}^S) = \lambda \mathcal{P}_\Omega(\text{sgn}(\mathcal{E}_0)). \tag{D.3}$$

However, $\mathcal{W}^S$ must reside in $\mathcal{T}^\perp$, making this a constrained linear equation.

Definition 2 (Neumann Series Construction)**.** *Define $\mathcal{W}^S$ via the following Neumann series:*

$$\mathcal{W}^S := \lambda \mathcal{P}_{\mathcal{T}^\perp} \sum_{k=0}^{\infty} (\mathcal{P}_\Omega \mathcal{P}_\mathcal{T} \mathcal{P}_\Omega)^k (sgn(\mathcal{E}_0) \times_3 D_0^T), \tag{D.4}$$

or equivalently:

$$\mathcal{W}^S = \lambda \mathcal{P}_{\mathcal{T}^\perp} (\mathcal{I} - \mathcal{P}_\Omega \mathcal{P}_\mathcal{T} \mathcal{P}_\Omega)^{-1} \mathcal{Z}. \tag{D.5}$$

By definition, $\mathcal{W}^S = \mathcal{P}_{\mathcal{T}^\perp}(\mathcal{I} - \mathcal{P}_\Omega \mathcal{P}_\mathcal{T} \mathcal{P}_\Omega)^{-1} \mathcal{Z}$, thus $\mathcal{W}^S \in \mathcal{T}^\perp$. Algebraic verification shows:

$$\mathcal{P}_\Omega(\mathcal{W}^S) = \lambda \mathcal{P}_\Omega(\mathcal{I} - \mathcal{P}_\mathcal{T})(\mathcal{I} - \mathcal{P}_\Omega \mathcal{P}_\mathcal{T} \mathcal{P}_\Omega)^{-1} \mathcal{Z} = \lambda \mathcal{P}_\Omega(\mathcal{Z}). \tag{D.6}$$

Appendix D.2. Convergence and Operator Norm Bound

The convergence of the Neumann series depends on the norm of the key operator $\mathcal{P}_\Omega \mathcal{P}_\mathcal{T} \mathcal{P}_\Omega$.

Lemma 9 (Operator Norm Bound)**.** *Under the Modified Adjusted Tensor Incoherence Condition (Definition 1), if the sparsity satisfies:*

$$\rho_s \leq \frac{c_0}{\mu \mu_D R r_3 \log^2(n_{(1)})}, \tag{D.7}$$

then with high probability ($\geq 1 - 2n_{(1)}^{-10}$):

$$\|\mathcal{P}_\Omega \mathcal{P}_\mathcal{T} \mathcal{P}_\Omega\| \leq \frac{1}{2}. \tag{D.8}$$

Proof. According to the tangent space definition of t-SVD, $\mathcal{P}_\mathcal{T} = \mathcal{P}_\mathcal{U} + \mathcal{P}_\mathcal{V} - \mathcal{P}_\mathcal{U}\mathcal{P}_\mathcal{V}$, where:

$$\mathcal{P}_\mathcal{U}(\mathcal{X}) = \mathcal{U} *_D \mathcal{U}^* *_D \mathcal{X}, \tag{D.9}$$
$$\mathcal{P}_\mathcal{V}(\mathcal{X}) = \mathcal{X} *_D \mathcal{V} *_D \mathcal{V}^*. \tag{D.10}$$

Thus:

$$\|\mathcal{P}_\Omega \mathcal{P}_\mathcal{T} \mathcal{P}_\Omega\| \leq \|\mathcal{P}_\Omega \mathcal{P}_\mathcal{U} \mathcal{P}_\Omega\| + \|\mathcal{P}_\Omega \mathcal{P}_\mathcal{V} \mathcal{P}_\Omega\| + \|\mathcal{P}_\Omega \mathcal{P}_\mathcal{U} \mathcal{P}_\mathcal{V} \mathcal{P}_\Omega\|. \tag{D.11}$$

Recall Lemma 3 (Basis tensor projection bound of Golfing Scheme): for basis tensor $\mathcal{E}_{ijk} = e_i \otimes e_j \otimes e_k^{(\text{orig})} \times_3 D_0^T$,

$$\|\mathcal{P}_\mathcal{T}(\mathcal{E}_{ijk})\|_F^2 \leq \frac{\mu \mu_D R r_3}{n_1 n_2 n_3}. \tag{D.12}$$

This implies that for any tensor $\mathcal{X}$:

$$\mathbb{E}[\|\mathcal{P}_\Omega \mathcal{P}_\mathcal{T}(\mathcal{X})\|_F^2] \leq \rho_s \cdot \frac{\mu \mu_D R r_3}{n_{(1)} n_3} \|\mathcal{X}\|_F^2. \tag{D.13}$$

Using Bernstein inequality, the operator norm satisfies:

$$\|\mathcal{P}_\Omega \mathcal{P}_\mathcal{T} \mathcal{P}_\Omega\| \leq C \rho_s \mu \mu_D R r_3 \log^2(n_{(1)}). \tag{D.14}$$

Choosing the constant $c_0 = 1/(2C)$ yields the required result. □

Appendix D.3. Norm Estimation

Lemma 10 (Norm Bound of Neumann Series)**.** *Under conditions of Lemma 9, the spectral norm of $\mathcal{W}^S$ satisfies:*

$$\|\mathcal{W}^S\| \leq 2\lambda\|\mathcal{Z}\|. \tag{D.15}$$

Proof. By the definition of Neumann series:

$$\begin{aligned}\|\mathcal{W}^S\| &= \left\|\lambda\mathcal{P}_{\mathcal{T}^\perp}\sum_{k=0}^{\infty}(\mathcal{P}_\Omega\mathcal{P}_\mathcal{T}\mathcal{P}_\Omega)^k\mathcal{Z}\right\| \\ &\leq \lambda\|\mathcal{P}_{\mathcal{T}^\perp}\|\cdot\left\|\sum_{k=0}^{\infty}(\mathcal{P}_\Omega\mathcal{P}_\mathcal{T}\mathcal{P}_\Omega)^k\right\|\cdot\|\mathcal{Z}\|. \end{aligned}\tag{D.16}$$

Since $\|\mathcal{P}_{\mathcal{T}^\perp}\| = 1$ (orthogonal projection) and $\|\mathcal{P}_\Omega\mathcal{P}_\mathcal{T}\mathcal{P}_\Omega\| \leq 1/2$, summing the geometric series:

$$\left\|\sum_{k=0}^{\infty}(\mathcal{P}_\Omega\mathcal{P}_\mathcal{T}\mathcal{P}_\Omega)^k\right\| \leq \frac{1}{1-1/2} = 2. \tag{D.17}$$

Thus:

$$\|\mathcal{W}^S\| \leq 2\lambda\|\mathcal{Z}\|. \tag{D.18}$$

□

The bound in Lemma 10 depends on the spectral norm of the target tensor $\mathcal{Z}$. We dedicate the following section to precisely estimating this quantity for a random sparse tensor.

Appendix D.4. Key Estimation of Spectral Norm of Sparse Tensor

To control $\|\mathcal{Z}\|$, we need the following core lemma.

Lemma 11 (Spectral Norm of Sparse Tensor under Tucker Projection)**.** *Let $\mathcal{M}_0 \in \mathbb{R}^{n_1\times n_2\times n_3}$ be a Bernoulli random sparse tensor:*

$$(\mathcal{M}_0)_{ijk} = \begin{cases}\pm 1, & (i,j,k)\in\Omega,\ \textit{taking each with prob. } \rho_s/2 \\ 0, & \textit{with prob. } 1-\rho_s\end{cases} \tag{D.19}$$

where locations are i.i.d. Let $D_0 \in \mathbb{R}^{n_3\times r_3}$ satisfy:

1. *Column orthogonality:* $D_0^T D_0 = I_{r_3}$;

2. *Row incoherence:* $\max_{\ell \in [n_3]} \|D_0(\ell, :)\|^2 \leq \frac{\mu_D r_3}{n_3}$.

Define the projected tensor $\mathcal{M} = \mathcal{M}_0 \times_3 D_0^T \in \mathbb{R}^{n_1 \times n_2 \times r_3}$. *Then with probability at least* $1 - 2(n_{(1)} r_3)^{-10}$:

$$\|\mathcal{M}\| \leq C\sqrt{\mu_D \rho_s n_{(1)} r_3}, \tag{D.20}$$

where $C > 0$ *is a universal constant and* $n_{(1)} = \max\{n_1, n_2\}$.

Proof. According to standard definitions in the t-SVD framework [9], the spectral norm of tensor $\mathcal{M} \in \mathbb{R}^{n_1 \times n_2 \times r_3}$ is the maximum spectral norm of its frequency slices:

$$\|\mathcal{M}\| = \|\overline{\mathcal{M}}\| = \max_{k \in [r_3]} \|\overline{\mathcal{M}}^{(k)}\|, \tag{D.21}$$

where $\overline{\mathcal{M}}$ is the DFT of $\mathcal{M}$ along the third dimension, and $\overline{\mathcal{M}}^{(k)} \in \mathbb{R}^{n_1 \times n_2}$ is the k-th frequency slice.

By the definition of Mode-3 product:

$$\mathcal{M}(i, j, s) = \sum_{\ell=1}^{n_3} \mathcal{M}_0(i, j, \ell) D_0^T(s, \ell) = \sum_{\ell=1}^{n_3} \mathcal{M}_0(i, j, \ell) D_0(\ell, s). \tag{D.22}$$

Applying the DFT along the third dimension:

$$\begin{aligned}
\overline{\mathcal{M}}^{(k)}(i, j) &= \frac{1}{\sqrt{r_3}} \sum_{s=1}^{r_3} \mathcal{M}(i, j, s) \omega_{r_3}^{-ks} \\
&= \frac{1}{\sqrt{r_3}} \sum_{s=1}^{r_3} \sum_{\ell=1}^{n_3} \mathcal{M}_0(i, j, \ell) D_0(\ell, s) \omega_{r_3}^{-ks} \\
&= \sum_{\ell=1}^{n_3} \mathcal{M}_0(i, j, \ell) \underbrace{\left(\frac{1}{\sqrt{r_3}} \sum_{s=1}^{r_3} D_0(\ell, s) \omega_{r_3}^{-ks} \right)}_{=: \tilde{D}_0^{(k)}(\ell)},
\end{aligned} \tag{D.23}$$

where $\omega_{r_3} = e^{2\pi i / r_3}$ and $\tilde{D}_0^{(k)}(\ell)$ is the k-th frequency component of the DFT of the ℓ-th row of D_0. Thus:

$$\overline{\mathcal{M}}^{(k)}(i, j) = \sum_{\ell=1}^{n_3} \mathcal{M}_0(i, j, \ell) \tilde{D}_0^{(k)}(\ell). \tag{D.24}$$

This can be written as a weighted sum:

$$\overline{\mathcal{M}}^{(k)} = \sum_{\ell=1}^{n_3} \tilde{D}_0^{(k)}(\ell) \cdot \mathcal{M}_0^{(\ell)}, \tag{D.25}$$

where $\mathcal{M}_0^{(\ell)} \in \mathbb{R}^{n_1 \times n_2}$ is the ℓ-th frontal slice of $\mathcal{M}_0$.

By Parseval's theorem and the column orthogonality of D_0:

$$\sum_{k=1}^{r_3} |\tilde{D}_0^{(k)}(\ell)|^2 = \|D_0(\ell, :)\|^2 \leq \frac{\mu_D r_3}{n_3}. \tag{D.26}$$

In particular, there exists at least one k such that $|\tilde{D}_0^{(k)}(\ell)| \leq \sqrt{\frac{\mu_D r_3}{n_3}}$. However, we need to control the contribution across all ℓ for each k.

For fixed $k \in [r_3]$:

$$\|\overline{\mathcal{M}}^{(k)}\| = \sup_{\substack{\|x\|=1, x \in \mathbb{R}^{n_1} \\ \|y\|=1, y \in \mathbb{R}^{n_2}}} \left\langle x, \sum_{\ell=1}^{n_3} \tilde{D}_0^{(k)}(\ell) \mathcal{M}_0^{(\ell)} y \right\rangle. \tag{D.27}$$

Construct 1/4-covering nets:

- $\mathcal{N}_1 \subset S^{n_1-1}$ with size $|\mathcal{N}_1| \leq 9^{n_1}$;
- $\mathcal{N}_2 \subset S^{n_2-1}$ with size $|\mathcal{N}_2| \leq 9^{n_2}$.

By standard net properties [7]:

$$\|\overline{\mathcal{M}}^{(k)}\| \leq 2 \max_{x' \in \mathcal{N}_1, y' \in \mathcal{N}_2} \left| \sum_{\ell=1}^{n_3} \tilde{D}_0^{(k)}(\ell) \langle x', \mathcal{M}_0^{(\ell)} y' \rangle \right|. \tag{D.28}$$

For fixed (x, y, k), define:

$$X_{x,y,k} := \sum_{\ell=1}^{n_3} \tilde{D}_0^{(k)}(\ell) \sum_{i=1}^{n_1} \sum_{j=1}^{n_2} x_i (\mathcal{M}_0)_{ij\ell} y_j. \tag{D.29}$$

Let $\xi_{ij\ell} := (\mathcal{M}_0)_{ij\ell}$ and $a_{ij\ell} := \tilde{D}_0^{(k)}(\ell) x_i y_j$. Then:

$$X_{x,y,k} = \sum_{i,j,\ell} \xi_{ij\ell} a_{ij\ell}. \tag{D.30}$$

Since $\mathbb{E}[\xi_{ij\ell}] = 0$ and $\mathbb{E}[\xi_{ij\ell}^2] = \rho_s$ (as $\xi_{ij\ell} \in \{0, \pm 1\}$ independently):

$$\begin{aligned}
\mathrm{Var}(X_{x,y,k}) &= \sum_{i,j,\ell} |a_{ij\ell}|^2 \rho_s \\
&= \rho_s \sum_{i,j,\ell} |\tilde{D}_0^{(k)}(\ell)|^2 |x_i|^2 |y_j|^2 \\
&= \rho_s \sum_{\ell=1}^{n_3} |\tilde{D}_0^{(k)}(\ell)|^2 \sum_{i=1}^{n_1} |x_i|^2 \sum_{j=1}^{n_2} |y_j|^2 \\
&= \rho_s \sum_{\ell=1}^{n_3} |\tilde{D}_0^{(k)}(\ell)|^2 \cdot 1 \cdot 1. \qquad \text{(D.31)}
\end{aligned}$$

Using Parseval's theorem:

$$\sum_{\ell=1}^{n_3} |\tilde{D}_0^{(k)}(\ell)|^2 \leq \sum_{\ell=1}^{n_3} \frac{\mu_D r_3}{n_3} = \mu_D r_3. \qquad \text{(D.32)}$$

Thus:

$$\mathrm{Var}(X_{x,y,k}) \leq \rho_s \mu_D r_3. \qquad \text{(D.33)}$$

Since $|\xi_{ij\ell}| \leq 1$ and $|a_{ij\ell}| \leq |\tilde{D}_0^{(k)}(\ell)| \leq \sqrt{\mu_D r_3 / n_3}$, using Bernstein inequality:

$$\mathbb{P}[|X_{x,y,k}| \geq t] \leq 2 \exp\left(-\frac{t^2/2}{\mathrm{Var}(X_{x,y,k}) + \max_{ij\ell} |a_{ij\ell}| t/3} \right). \qquad \text{(D.34)}$$

Taking $t = C_0 \sqrt{\rho_s \mu_D r_3 \log(n_{(1)} r_3)}$ (with C_0 large enough) and assuming $t \gg \max |a_{ij\ell}| \sim \sqrt{\mu_D r_3 / n_3}$ (which holds for large n_3):

$$\mathbb{P}[|X_{x,y,k}| \geq t] \leq 2 \exp\left(-\frac{C_0^2 \log(n_{(1)} r_3)}{2} \right) \leq 2 (n_{(1)} r_3)^{-11}. \qquad \text{(D.35)}$$

Applying Union Bound over all $(x, y, k) \in \mathcal{N}_1 \times \mathcal{N}_2 \times [r_3]$. Note that:

$$\begin{aligned}
|\mathcal{N}_1| \cdot |\mathcal{N}_2| \cdot r_3 &\leq 9^{n_1} \cdot 9^{n_2} \cdot r_3 \\
&\leq 9^{2n_{(1)}} \cdot r_3 \\
&\leq \exp(2 n_{(1)} \log 9) \cdot r_3 \\
&\leq (n_{(1)} r_3)^{10} \quad \text{(for sufficiently large } n_{(1)}). \qquad \text{(D.36)}
\end{aligned}$$

The failure probability is bounded by:

$$\leq (n_{(1)}r_3)^{10} \cdot 2(n_{(1)}r_3)^{-11} = 2(n_{(1)}r_3)^{-1} \leq 2(n_{(1)}r_3)^{-10}. \tag{D.37}$$

By net properties:

$$\|\overline{\mathcal{M}}^{(k)}\| \leq 2 \max_{x' \in \mathcal{N}_1, y' \in \mathcal{N}_2} |X_{x',y',k}| \leq 2t = 2C_0\sqrt{\rho_s \mu_D r_3 \log(n_{(1)}r_3)}. \tag{D.38}$$

Taking the maximum over all k and absorbing the logarithmic factors into the constant:

$$\|\mathcal{M}\| = \max_{k \in [r_3]} \|\overline{\mathcal{M}}^{(k)}\| \leq C\sqrt{\mu_D \rho_s n_{(1)} r_3}. \tag{D.39}$$

□

Applying Lemma 11 to $\mathcal{Z} = \lambda \text{sgn}(\mathcal{E}_0) \times_3 D_0^T$ with $M_0 = \text{sgn}(\mathcal{E}_0)$ and $D = D_0$. Note $n_{(1)} = \max\{n_1, n_2\}$:

$$||\mathcal{Z}|| = \lambda ||\mathcal{M}\|| \leq \lambda C\sqrt{\mu_D \rho_s n_{(1)} r_3}. \tag{D.40}$$

Substituting $\lambda = 1/\sqrt{n_{(1)}}$, the $\sqrt{n_{(1)}}$ terms cancel out:

$$||\mathcal{Z}|| \leq C\sqrt{\mu_D \rho_s r_3}. \tag{D.41}$$

From the norm estimation of $\mathcal{W}^S$:

$$||\mathcal{W}^S|| \leq 2||\mathcal{Z}|| \leq 2C\sqrt{\mu_D \rho_s r_3}. \tag{D.42}$$

Since $\mathcal{W}^S$ is intended to compensate for residue $\mathcal{R}$ (primarily on $T^\perp$), we have:

$$||P_{T^\perp}(\mathcal{W}^S)|| \leq ||\mathcal{W}^S|| \leq 2C\sqrt{\mu_D \rho_s r_3}. \tag{D.43}$$

We now verify whether the overall dual certificate $\mathcal{W} = \mathcal{W}^L + \mathcal{W}^S$ satisfies the norm bound condition in Lemma 1: $||\mathcal{W}|| \leq 1/4$.

$$||\mathcal{W}|| \leq ||\mathcal{W}^L|| + ||\mathcal{W}^S||. \tag{D.44}$$

From the conclusion of Lemma 2, for sufficiently large iteration count j_0, we have $||\mathcal{W}^L|| \leq n_{(1)}^{-10}$. This is a negligible term for large $n_{(1)}$. Using the result:

$$||\mathcal{W}^S|| \leq 2C\sqrt{\mu_D \rho_s r_3}. \tag{D.45}$$

To satisfy $||\mathcal{W}|| \leq 1/4$, we need:

$$2C\sqrt{\mu_D \rho_s r_3} \leq \frac{1}{4}. \tag{D.46}$$

This directly leads to the sparsity condition (Condition 3) in the main theorem:

$$\rho_s \leq \frac{1}{64C^2 \mu_D r_3} \implies \rho_s \leq \frac{c_0}{\mu_D r_3}. \tag{D.47}$$

In summary, when Condition 3 of Theorem 1 ($\rho_s \leq \frac{c_0}{\mu_D r_3}$) holds, we have $\|\mathcal{W}^S\| \leq \frac{1}{8}$. Combined with the result $\|\mathcal{W}^L\| \leq \frac{1}{8}$ from Lemma 2, we get $\mathcal{W} = \mathcal{W}^L + \mathcal{W}^S$ satisfying $\|\mathcal{W}\| \leq \|\mathcal{W}^L\| + \|\mathcal{W}^S\| \leq \frac{1}{4}$, satisfying Condition 1 of Lemma 1. This completes the construction and verification of $\mathcal{W}^S$.

Appendix E. Verification of Dual Proof Conditions

Finally, having constructed $\mathcal{W}^L$ via the Golfing Scheme and $\mathcal{W}^S$ via the Neumann series, we must verify that their composite sum strictly satisfies all the conditions for the dual certificate stipulated in Lemma 1.

Lemma 12 (Verification of Dual Proof Conditions)**.** *Let $\mathcal{W} = \mathcal{W}^L + \mathcal{W}^S$ (constructed via Lemma 2 and Definition 2), and define $\mathcal{F}$ and $\mathcal{D}$ as follows:*

$$\mathcal{F} = -\frac{1}{\lambda}\mathcal{P}_{\Omega^\perp}\mathcal{W}^S, \tag{E.1}$$

$$\mathcal{D} = \frac{1}{\lambda}\mathcal{P}_{\Omega^\perp}(\omega(\vartheta \otimes (U *_D V^* + \mathcal{W}^L)) \times_3 D_0 - \mathrm{sgn}(\mathcal{E}_0)). \tag{E.2}$$

Then with high probability (under assumptions of Theorem 1), $(\mathcal{W}, \mathcal{F}, \mathcal{D})$ satisfies all conditions of Lemma 1.

Proof. By construction: $\mathcal{W}^L = \mathcal{P}_{\mathcal{T}^\perp} Y_{j_0}$, so $\mathcal{P}_{\mathcal{T}}\mathcal{W}^L = 0$.

$$\mathcal{W}^S = \lambda \mathcal{P}_{\mathcal{T}^\perp}(\mathcal{I} - \mathcal{P}_\Omega \mathcal{P}_{\mathcal{T}} \mathcal{P}_\Omega)^{-1}\mathcal{Z}. \tag{E.3}$$

Thus $\mathcal{P}_{\mathcal{T}}\mathcal{W}^S = 0$. Hence $\mathcal{P}_{\mathcal{T}}\mathcal{W} = \mathcal{P}_{\mathcal{T}}(\mathcal{W}^L + \mathcal{W}^S) = 0$.

From Lemma 2 and results of Section Appendix D:

$$\|\mathcal{W}^L\| \leq C_1 \epsilon \log(n_{(1)}) \sqrt{r r_3} \leq \frac{1}{8}, \tag{E.4}$$

$$\|\mathcal{W}^S\| \leq 2C \sqrt{\mu_D \rho_s r_3} \leq \frac{1}{8}. \tag{E.5}$$

Thus:

$$\|\mathcal{W}\| \leq \|\mathcal{W}^L\| + \|\mathcal{W}^S\| \leq \frac{1}{8} + \frac{1}{8} = \frac{1}{4}. \tag{E.6}$$

$\mathcal{F} = -\frac{1}{\lambda} \mathcal{P}_{\Omega^\perp} \mathcal{W}^S$, so:

$$\mathcal{P}_\Omega \mathcal{F} = -\frac{1}{\lambda} \mathcal{P}_\Omega \mathcal{P}_{\Omega^\perp} \mathcal{W}^S = 0. \tag{E.7}$$

From the properties established in Section Appendix D:

$$\|\mathcal{F}\|_\infty = \frac{1}{\lambda} \|\mathcal{P}_{\Omega^\perp} \mathcal{W}^S\|_\infty \leq \frac{1}{\lambda} \cdot \frac{\lambda}{8} = \frac{1}{8} < \frac{1}{2}. \tag{E.8}$$

From the Dual Proof condition:

$$\omega(\vartheta \otimes (U *_D V^* + \mathcal{W})) = \lambda[(\operatorname{sgn}(\mathcal{E}_0) + \mathcal{F} + \mathcal{P}_{\Omega^\perp} \mathcal{D}) \times_3 D_0^{\mathrm{T}}]. \tag{E.9}$$

We define $\mathcal{D}$ such that this condition holds. Multiply by $\times_3 D_0$ on both sides:

$$\omega(\vartheta \otimes (U *_D V^* + \mathcal{W})) \times_3 D_0 = \lambda(\operatorname{sgn}(\mathcal{E}_0) + \mathcal{F} + \mathcal{P}_{\Omega^\perp} \mathcal{D}). \tag{E.10}$$

Apply $\mathcal{P}_\Omega$ projection and note $\mathcal{P}_\Omega \mathcal{F} = 0$:

$$\mathcal{P}_\Omega[\omega(\vartheta \otimes (U *_D V^* + \mathcal{W})) \times_3 D_0] = \lambda \mathcal{P}_\Omega \operatorname{sgn}(\mathcal{E}_0) + \lambda \mathcal{P}_\Omega \mathcal{P}_{\Omega^\perp} \mathcal{D}. \tag{E.11}$$

Since $\mathcal{P}_\Omega \mathcal{P}_{\Omega^\perp} = 0$, the second term vanishes. Thus we focus on $\mathcal{P}_{\Omega^\perp}$ part:

$$\mathcal{P}_{\Omega^\perp} \mathcal{D} = \frac{1}{\lambda} \mathcal{P}_{\Omega^\perp}[\omega(\vartheta \otimes (U *_D V^* + \mathcal{W})) \times_3 D_0 - \operatorname{sgn}(\mathcal{E}_0) - \mathcal{F}]. \tag{E.12}$$

Since $\mathcal{F}$ is supported entirely on $\Omega^\perp$, this simplifies to directly defining $\mathcal{D}$. Based on the properties of dual conditions, define:

$$\mathcal{D} := \frac{1}{\lambda} \mathcal{P}_{\Omega^\perp}[\omega(\vartheta \otimes (U *_D V^* + \mathcal{W})) \times_3 D_0 - \operatorname{sgn}(\mathcal{E}_0)]. \tag{E.13}$$

Substitute $\mathcal{W} = \mathcal{W}^L + \mathcal{W}^S$ into the above:

$$\begin{aligned}
\mathcal{D} &= \frac{\omega}{\lambda}\mathcal{P}_{\Omega^\perp}[(\vartheta \otimes (U *_D V^* + \mathcal{W}^L + \mathcal{W}^S)) \times_3 D_0] - \frac{1}{\lambda}\mathcal{P}_{\Omega^\perp}\operatorname{sgn}(\mathcal{E}_0) \\
&= \frac{\omega}{\lambda}\mathcal{P}_{\Omega^\perp}[(\vartheta \otimes (U *_D V^* + \mathcal{W}^L)) \times_3 D_0] \\
&\quad + \frac{\omega}{\lambda}\mathcal{P}_{\Omega^\perp}[(\vartheta \otimes \mathcal{W}^S) \times_3 D_0] \\
&\quad - \frac{1}{\lambda}\mathcal{P}_{\Omega^\perp}\operatorname{sgn}(\mathcal{E}_0). \qquad \text{(E.14)}
\end{aligned}$$

We decompose this into three parts:

$$\mathcal{D}_1 = \frac{\omega}{\lambda}\mathcal{P}_{\Omega^\perp}[(\vartheta \otimes (U *_D V^* + \mathcal{W}^L)) \times_3 D_0], \qquad \text{(E.15)}$$

$$\mathcal{D}_2 = \frac{\omega}{\lambda}\mathcal{P}_{\Omega^\perp}[(\vartheta \otimes \mathcal{W}^S) \times_3 D_0], \qquad \text{(E.16)}$$

$$\mathcal{D}_3 = -\frac{1}{\lambda}\mathcal{P}_{\Omega^\perp}\operatorname{sgn}(\mathcal{E}_0). \qquad \text{(E.17)}$$

Then $\mathcal{D} = \mathcal{D}_1 + \mathcal{D}_2 + \mathcal{D}_3$.

We use the low-rankness and spectral norm bounds of $\mathcal{W}^L$ constructed by the Golfing Scheme.

1. From Golfing Scheme, $\mathcal{W}^L = \sum_{j=1}^{k_0} Y_j$. Since $Y_j \in \mathcal{T}$ and the tubal rank in $\mathcal{T}$ does not exceed $2r$, the tubal rank of $\mathcal{W}^L$ is bounded by $2r$. Thus $\operatorname{rank}_{tubal}((\vartheta \otimes \mathcal{W}^L) \times_3 D_0) \leq 2r$.

2. Using Bernstein inequality, the dual certificate converges in spectral norm. Constant C_1 exists such that:

$$\|(\vartheta \otimes \mathcal{W}^L) \times_3 D_0\| \leq C_1 \vartheta_{\max}. \qquad \text{(E.18)}$$

3. Using the norm equivalence $\|\mathcal{A}\|_F \leq \sqrt{r_3 \cdot \operatorname{rank}(\mathcal{A})}\|\mathcal{A}\|$:

$$\begin{aligned}
\|\mathcal{D}_1\|_F &= \frac{\omega}{\lambda}\|\mathcal{P}_{\Omega^\perp}[(\vartheta \otimes \mathcal{W}^L) \times_3 D_0]\|_F \\
&\leq \frac{\omega}{\lambda}\|(\vartheta \otimes \mathcal{W}^L) \times_3 D_0\|_F \\
&\leq \frac{\omega}{\lambda}\sqrt{r_3(2r)}\|(\vartheta \otimes \mathcal{W}^L) \times_3 D_0\| \\
&\leq \frac{\omega}{\lambda}\sqrt{2rr_3}(C_1\vartheta_{\max}). \qquad \text{(E.19)}
\end{aligned}$$

We utilize the random sign property of $\mathcal{E}_0$ to eliminate the dependence on n.

1. $\mathcal{D}_2 = \frac{\omega}{\lambda}\mathcal{P}_{\Omega^\perp}[(\vartheta \otimes \mathcal{W}^S) \times_3 D_0]$, where $\mathcal{W}^S = \mathcal{P}_{\mathcal{T}}(\mathcal{Z})$ is the projection of $\mathcal{Z} = \lambda \text{sgn}(\mathcal{E}_0) \times_3 D_0^T$ onto the tangent space.

2. For random sign tensor $\text{sgn}(\mathcal{E}_0)$, spectral norm satisfies with high probability:
$$\|\text{sgn}(\mathcal{E}_0)\| \leq C_0\sqrt{\rho_s n_{(1)} \log(n_{(1)})}, \tag{E.20}$$
where ρ_s is the sparsity ratio and C_0 is an absolute constant. Hence, the spectral norm of $\mathcal{Z}$ is bounded by:
$$\begin{aligned}\|\mathcal{Z}\| &= \|\lambda \text{sgn}(\mathcal{E}_0) \times_3 D_0^T\| \\ &\leq \lambda \|\text{sgn}(\mathcal{E}_0)\| \\ &\leq \frac{1}{\sqrt{n_{(1)}}} \cdot C_0\sqrt{\rho_s n_{(1)} \log n_{(1)}} = C_0\sqrt{\rho_s \log n_{(1)}}.\end{aligned} \tag{E.21}$$

3. Since $\mathcal{W}^S \in \mathcal{T}$, its tubal rank $\leq 2r$. Tangent space projection satisfies $\|\mathcal{W}^S\| \leq 2\|\mathcal{Z}\|$.
$$\begin{aligned}\|\mathcal{W}^S\|_F &\leq \sqrt{2rr_3}\|\mathcal{W}^S\| \\ &\leq \sqrt{2rr_3} \cdot 2\|\mathcal{Z}\| \\ &\leq 2C_0\sqrt{2rr_3\rho_s \log n_{(1)}}.\end{aligned} \tag{E.22}$$

4. Substitute into $\mathcal{D}_2$:
$$\begin{aligned}\|\mathcal{D}_2\|_F &\leq \frac{\omega}{\lambda}\vartheta_{\max}\|\mathcal{W}^S\|_F \\ &\leq \frac{\omega}{\lambda}\vartheta_{\max}\left(2C_0\sqrt{2rr_3\rho_s \log n_{(1)}}\right).\end{aligned} \tag{E.23}$$

   Combining (E.19) and (E.23), if ω is same order as λ ($\omega/\lambda \approx 1$), then small enough ρ_s and r, r_3 ensure $\|\mathcal{D}_1\|_F + \|\mathcal{D}_2\|_F \leq 1/4$.

Note that the support of $\text{sgn}(\mathcal{E}_0)$ lies entirely within Ω; thus, its projection onto $\Omega^\perp$ is 0. Thus:
$$||\mathcal{D}_3||_F = \frac{1}{\lambda}||\mathcal{P}_{\Omega^\perp}\text{sgn}(\mathcal{E}_0)||_F = 0. \tag{E.24}$$

Combine the bounds:

$$\|\mathcal{D}\|_F \leq \frac{\omega}{\lambda}\vartheta_{\max}\sqrt{2rr_3}\left(C_1 + 2C_0\sqrt{\rho_s \log n_{(1)}}\right). \tag{E.25}$$

To satisfy $\|\mathcal{P}_{\Omega^\perp}\mathcal{D}\|_F \leq \frac{\sqrt{r_3}}{4}$ (as $\sqrt{r_3} \geq 1$, we verify the stronger $\|\mathcal{D}\|_F \leq 1/4$):

$$\frac{\omega}{\lambda}\vartheta_{\max}\sqrt{rr_3}\left(\sqrt{2}C_1 + 2\sqrt{2}C_0\sqrt{\rho_s \log n_{(1)}}\right) \leq \frac{1}{4}. \tag{E.26}$$

Analyzing the parameters, we have:

1. If $\omega = c\lambda$, then $\frac{\omega}{\lambda}$ is constant.
2. This holds provided that ρ_s is sufficiently small (Theorem 1, Condition 3).
3. For low-rank tensors, these are much smaller than the tensor dimensions.

Therefore, for sufficiently low-rank and sparse noise, (E.26) holds. This completes the verification of dual certificate $\mathcal{D}$.

□

Appendix F. Algorithm Convergence Analysis

In this section, we establish the theoretical convergence guarantees for the Alternating Direction Method of Multipliers (ADMM) framework used to solve the AC^2-TRPCA model. Since the proposed model introduces a convex-concave separation function and involves optimization on a Stiefel manifold (orthogonal constraints), traditional convex optimization convergence theory does not directly apply. To verify the numerical stability and reliability of the algorithm, we provide a rigorous theoretical convergence analysis based on variational analysis and limiting subdifferential theory.

Specifically, we first prove the boundedness of the algorithm's iteration sequence, then establish the convergence of the primal feasibility residual, and finally demonstrate that the iteration sequence converges to a generalized Karush-Kuhn-Tucker (KKT) point of the original nonconvex optimization problem.

Appendix F.1. Augmented Lagrangian and Iteration Rules

For completeness, we first restate the augmented Lagrangian function and the iteration rules of our algorithm. The augmented Lagrangian function for the AC^2-TRPCA model is defined as:

$$\begin{aligned}\mathcal{L}(\mathcal{B}, D, \mathcal{E}, \Lambda, \mu) = \mathrm{Sep}(\mathcal{B}, \Omega, \vartheta) + \lambda\|\mathcal{E}\|_1 \\ + \langle \Lambda, \mathcal{Y} - \mathcal{B} \times_3 D - \mathcal{E}\rangle + \frac{\mu}{2}\|\mathcal{Y} - \mathcal{B} \times_3 D - \mathcal{E}\|_F^2.\end{aligned} \tag{F.1}$$

At the k-th iteration, the algorithm sequentially updates the variables as follows:

$$\mathcal{B}^{(k+1)} = \arg\min_{\mathcal{B}} \mathrm{Sep}(\mathcal{B}, \Omega, \vartheta) + \frac{\mu^{(k)}}{2}\left\|\mathcal{B} - \left(\mathcal{Y} - \mathcal{E}^{(k)} + \frac{\Lambda^{(k)}}{\mu^{(k)}}\right) \times_3 (D^{(k)})^T\right\|_F^2, \tag{F.2}$$

$$D^{(k+1)} = \arg\max_{D^T D = I_{r_3}} \mathrm{tr}\left(D^T M^{(k)}\right), \tag{F.3}$$

$$\mathcal{E}^{(k+1)} = \arg\min_{\mathcal{E}} \lambda\|\mathcal{E}\|_1 + \frac{\mu^{(k)}}{2}\left\|\mathcal{Y} - \mathcal{B}^{(k+1)} \times_3 D^{(k+1)} + \frac{\Lambda^{(k)}}{\mu^{(k)}} - \mathcal{E}\right\|_F^2, \tag{F.4}$$

$$\Lambda^{(k+1)} = \Lambda^{(k)} + \mu^{(k)}\left(\mathcal{Y} - \mathcal{B}^{(k+1)} \times_3 D^{(k+1)} - \mathcal{E}^{(k+1)}\right), \tag{F.5}$$

where $M^{(k)} = [\mathrm{unfold}_3\left(\mathcal{Y} - \mathcal{E}^{(k)} + \Lambda^{(k)}/\mu^{(k)}\right)]^T \cdot \mathrm{unfold}_3(\mathcal{B}^{(k+1)})$.

Appendix F.2. Preliminaries for Convergence

Before presenting the main theorems, we provide the basic definitions and lemmas from variational analysis needed for the proofs.

Definition 3 (Limiting Subdifferential [10])**.** *Let $G : \mathbb{R}^{n_1 \times n_2 \times n_3} \to \mathbb{R}$ be a proper lower semi-continuous function. The limiting subdifferential of G at $\mathcal{X}$ is defined as:*

$$\partial G(\mathcal{X}) = \left\{\mathcal{Z} : \exists \mathcal{X}^{(k)} \to \mathcal{X}, G(\mathcal{X}^{(k)}) \to G(\mathcal{X}), \mathcal{Z}^{(k)} \in \hat{\partial} G(\mathcal{X}^{(k)}) \to \mathcal{Z}\right\}, \tag{F.6}$$

where $\hat{\partial} G(\mathcal{X})$ is the Fréchet subdifferential.

Lemma 13 (Fermat's Rule [10])**.** *If $\mathcal{X}$ is a local minimum of G, then $0 \in \partial G(\mathcal{X})$.*

Lemma 14 (Sum Rule [10]). *If f is continuously differentiable and G is lower semi-continuous, then*

$$\partial(f+G)(\mathcal{X}) = \nabla f(\mathcal{X}) + \partial G(\mathcal{X}). \tag{F.7}$$

Lemma 15 (Outer Semicontinuity [10]). *Let $\mathcal{X}^{(k)} \to \mathcal{X}$, $G(\mathcal{X}^{(k)}) \to G(\mathcal{X})$, $\mathcal{Z}^{(k)} \in \partial G(\mathcal{X}^{(k)})$, and $\mathcal{Z}^{(k)} \to \mathcal{Z}$. Then $\mathcal{Z} \in \partial G(\mathcal{X})$.*

Appendix F.3. Proofs of Convergence Theorems

First, we prove that the sequence of Lagrange multipliers generated by the algorithm is bounded.

Theorem 2 (Boundedness of Lagrange Multipliers). *The sequence of Lagrange multipliers $\{\Lambda^{(k)}\}_{k\geq 1}$ generated by the algorithm is bounded. Specifically, for any $k \geq 1$:*

$$\|\Lambda^{(k)}\|_\infty \leq \lambda, \quad \textit{and} \quad \|\Lambda^{(k)}\|_F \leq \sqrt{n_1 n_2 n_3}\lambda. \tag{F.8}$$

Proof. From the first-order optimality condition (Fermat's Rule, Lemma 13) for the $\mathcal{E}$-subproblem in (F.4):

$$0 \in \lambda\partial\|\mathcal{E}^{(k+1)}\|_1 - \mu^{(k)}\left(\mathcal{Y} - \mathcal{B}^{(k+1)} \times_3 D^{(k+1)} - \mathcal{E}^{(k+1)} + \frac{\Lambda^{(k)}}{\mu^{(k)}}\right). \tag{F.9}$$

Rearranging yields:

$$\Lambda^{(k)} + \mu^{(k)}(\mathcal{Y} - \mathcal{B}^{(k+1)} \times_3 D^{(k+1)} - \mathcal{E}^{(k+1)}) \in \lambda\partial\|\mathcal{E}^{(k+1)}\|_1. \tag{F.10}$$

According to the multiplier update formula (F.5), the left-hand side is exactly $\Lambda^{(k+1)}$:

$$\Lambda^{(k+1)} \in \lambda\partial\|\mathcal{E}^{(k+1)}\|_1. \tag{F.11}$$

By the properties of the subdifferential of the ℓ_1-norm, for any $G \in \partial\|\mathcal{X}\|_1$, we have $\|G\|_\infty \leq 1$. Therefore:

$$\|\Lambda^{(k+1)}\|_\infty \leq \lambda. \tag{F.12}$$

By the equivalence of norms in finite-dimensional spaces:

$$\|\Lambda^{(k+1)}\|_F \leq \sqrt{n_1 n_2 n_3}\|\Lambda^{(k+1)}\|_\infty \leq \sqrt{n_1 n_2 n_3}\lambda =: C_0. \tag{F.13}$$

Hence, $\{\Lambda^{(k)}\}_{k\geq 1}$ is bounded. □

Building upon the boundedness of the multipliers, we next establish that the primal feasibility residual sequence converges to zero.

Theorem 3 (Convergence of Primal Residual)**.** *If the weights of the convex-concave separation function satisfy $\omega_1 \geq \omega_2 \geq \cdots \geq \omega_r \geq 0$, then the sequence generated by the algorithm ensures that the primal feasibility residual converges to zero:*

$$\lim_{k\to\infty} \|\mathcal{Y} - \mathcal{B}^{(k+1)} \times_3 D^{(k+1)} - \mathcal{E}^{(k+1)}\|_F = 0. \tag{F.14}$$

Proof. According to the conclusion of Theorem 2, the sequence $\{\Lambda^{(k)}\}_{k\geq 1}$ is bounded; that is, there exists a constant $C_0 > 0$ such that $\|\Lambda^{(k)}\|_F \leq C_0$ for any iteration k.

The multiplier update formula (F.5) in the algorithm is:

$$\Lambda^{(k+1)} = \Lambda^{(k)} + \mu^{(k)} \left(\mathcal{Y} - \mathcal{B}^{(k+1)} \times_3 D^{(k+1)} - \mathcal{E}^{(k+1)}\right). \tag{F.15}$$

Rearranging the equation to express the primal residual as a function of the change in multipliers:

$$\mathcal{Y} - \mathcal{B}^{(k+1)} \times_3 D^{(k+1)} - \mathcal{E}^{(k+1)} = \frac{1}{\mu^{(k)}} \left(\Lambda^{(k+1)} - \Lambda^{(k)}\right). \tag{F.16}$$

Taking the Frobenius norm on both sides and using the triangle inequality along with the boundedness of $\Lambda^{(k)}$:

$$\begin{aligned} \|\mathcal{Y} - \mathcal{B}^{(k+1)} \times_3 D^{(k+1)} - \mathcal{E}^{(k+1)}\|_F &= \frac{1}{\mu^{(k)}} \|\Lambda^{(k+1)} - \Lambda^{(k)}\|_F \\ &\leq \frac{1}{\mu^{(k)}} \left(\|\Lambda^{(k+1)}\|_F + \|\Lambda^{(k)}\|_F\right) \\ &\leq \frac{2C_0}{\mu^{(k)}}. \end{aligned} \tag{F.17}$$

Since the penalty parameter $\mu^{(k)}$ increases geometrically in the algorithm settings (i.e., $\mu^{(k+1)} = \min(\rho\mu^{(k)}, \mu_{\max})$ with $\rho > 1$), then $\mu^{(k)} \to +\infty$ as $k \to \infty$. Taking the limit as $k \to \infty$ on both sides, the right-hand side tends to 0, proving:

$$\lim_{k\to\infty} \|\mathcal{Y} - \mathcal{B}^{(k+1)} \times_3 D^{(k+1)} - \mathcal{E}^{(k+1)}\|_F = 0. \tag{F.18}$$

The proof is complete. □

Finally, we demonstrate that the iteration sequence converges to a generalized Karush-Kuhn-Tucker (KKT) point of the original optimization problem.

Theorem 4 (Convergence to KKT Point)**.** *The sequence* $\{(\mathcal{B}^{(k)}, D^{(k)}, \mathcal{E}^{(k)})\}$ *generated by the algorithm is bounded, and there exists a convergent subsequence. Any limit point* $(\mathcal{B}^*, D^*, \mathcal{E}^*, \Lambda^*)$ *of a convergent subsequence satisfies the generalized Karush-Kuhn-Tucker (KKT) conditions of the original optimization problem. Specifically, the limit point satisfies:*

$$\begin{cases} 0 \in \partial Sep(\mathcal{B}^*, \omega, \vartheta) - \Lambda^* \times_3 (D^*)^T, \\ 0 \in \lambda \partial \|\mathcal{E}^*\|_1 - \Lambda^*, \\ \mathcal{Y} = \mathcal{B}^* \times_3 D^* + \mathcal{E}^*, \\ (D^*)^T D^* = I_{r_3}, \\ D^* = Polar(M^*). \end{cases} \tag{F.19}$$

Proof. **(I) Sequence Boundedness and Subsequence Convergence**

First, from Theorem 2, the dual variable sequence $\{\Lambda^{(k)}\}_{k\geq 1}$ is bounded; there exists $C_0 > 0$ such that $\|\Lambda^{(k)}\|_F \leq C_0$ for all $k \geq 1$.

Second, utilizing the coercivity of the augmented Lagrangian function (F.1) and the optimality of the primal subproblems, we have the following descent inequality:

$$\mathcal{L}(\mathcal{B}^{(k+1)}, D^{(k+1)}, \mathcal{E}^{(k+1)}, \Lambda^{(k)}, \mu^{(k)}) \leq \mathcal{L}(\mathcal{B}^{(k)}, D^{(k)}, \mathcal{E}^{(k)}, \Lambda^{(k)}, \mu^{(k)}). \tag{F.20}$$

This indicates that at the k-th iteration, the primal variable update ensures that the objective function value is non-increasing. Since $\mathcal{L}$ is coercive (function value tends to infinity as variable norms tend to infinity) and the dual variables $\Lambda^{(k)}$ are bounded, we can infer that the primal variable sequence is also restricted to a compact region. Thus, there exists $C_1 > 0$ such that:

$$\|\mathcal{B}^{(k)}\|_F + \|\mathcal{E}^{(k)}\|_F \leq C_1, \quad \text{and} \quad D^{(k)} \in \mathcal{S}_{n_3,r_3}, \tag{F.21}$$

where $\mathcal{S}_{n_3,r_3} = \{D \in \mathbb{R}^{n_3 \times r_3} : D^T D = I_{r_3}\}$ denotes the Stiefel manifold.

Since the sequence $\mathcal{Z}^{(k)} := (\mathcal{B}^{(k)}, D^{(k)}, \mathcal{E}^{(k)}, \Lambda^{(k)})$ is uniformly bounded in a finite-dimensional space, by the Bolzano-Weierstrass Theorem, there must

exist a convergent subsequence $\{\mathcal{Z}^{(k_j)}\}_{j=1}^{\infty}$ converging to an accumulation point $\mathcal{Z}^*$:

$$\lim_{j\to\infty}(\mathcal{B}^{(k_j)}, D^{(k_j)}, \mathcal{E}^{(k_j)}, \Lambda^{(k_j)}) = (\mathcal{B}^*, D^*, \mathcal{E}^*, \Lambda^*). \tag{F.22}$$

In the following steps, we will prove that the limit point satisfies the KKT conditions.

(II) Stationarity with respect to $\mathcal{B}$

Step 1: First-order optimality condition of the B-subproblem The update rule for the $\mathcal{B}$-subproblem (F.2) at iteration k implies, by Fermat's Rule (Lemma 13):

$$0 \in \partial\left(\mathrm{Sep}(\mathcal{B}^{(k+1)}, \Omega, \vartheta) + \frac{\mu^{(k)}}{2}\left\|\mathcal{B}^{(k+1)} - \mathcal{N}^{(k)}\right\|_F^2\right), \tag{F.23}$$

where $\mathcal{N}^{(k)} = (\mathcal{Y} - \mathcal{E}^{(k)} + \frac{\Lambda^{(k)}}{\mu^{(k)}}) \times_3 (D^{(k)})^T$.

Step 2: Applying the subdifferential sum rule Since the quadratic term $\frac{\mu^{(k)}}{2}\|\mathcal{B}-\mathcal{N}^{(k)}\|_F^2$ is differentiable with respect to $\mathcal{B}$ with gradient $\mu^{(k)}(\mathcal{B}^{(k+1)} - \mathcal{N}^{(k)})$, applying the sum rule (Lemma 14):

$$0 \in \partial\mathrm{Sep}(\mathcal{B}^{(k+1)}, \Omega, \vartheta) + \mu^{(k)}\left(\mathcal{B}^{(k+1)} - \mathcal{N}^{(k)}\right). \tag{F.24}$$

Substituting the definition of $\mathcal{N}^{(k)}$:

$$0 \in \partial\mathrm{Sep}(\mathcal{B}^{(k+1)}, \Omega, \vartheta) + \mu^{(k)}\mathcal{B}^{(k+1)} - \mu^{(k)}\left(\mathcal{Y} - \mathcal{E}^{(k)} + \frac{\Lambda^{(k)}}{\mu^{(k)}}\right) \times_3 (D^{(k)})^T. \tag{F.25}$$

Step 3: Rearranging terms Rearranging to isolate the subgradient term:

$$\Lambda^{(k)} \times_3 (D^{(k)})^T \in \partial\mathrm{Sep}(\mathcal{B}^{(k+1)}, \Omega, \vartheta) + \mu^{(k)}\left[\mathcal{B}^{(k+1)} - (\mathcal{Y} - \mathcal{E}^{(k)}) \times_3 (D^{(k)})^T\right]. \tag{F.26}$$

Using the property $\mathcal{T} \times_3 (D^{(k)})^T \times_3 D^{(k)} = \mathcal{T}$ (since $(D^{(k)})^T D^{(k)} = I$), we rewrite the bracketed term:

$$\mathcal{B}^{(k+1)} - (\mathcal{Y} - \mathcal{E}^{(k)}) \times_3 (D^{(k)})^T = \left[\mathcal{B}^{(k+1)} \times_3 D^{(k)} + \mathcal{E}^{(k)} - \mathcal{Y}\right] \times_3 (D^{(k)})^T. \tag{F.27}$$

Thus, Eq. (F.26) becomes:

$$\begin{aligned}\Lambda^{(k)} \times_3 (D^{(k)})^T \in \partial\,\mathrm{Sep}(\mathcal{B}^{(k+1)}, \Omega, \vartheta) \\ + \mu^{(k)} \left[\mathcal{B}^{(k+1)} \times_3 D^{(k)} + \mathcal{E}^{(k)} - \mathcal{Y}\right] \times_3 (D^{(k)})^T.\end{aligned} \tag{F.28}$$

Step 4: Introducing variable updates and dual residuals Define the difference in variables between successive iterations as:

$$\Delta D^{(k)} = D^{(k+1)} - D^{(k)}, \quad \Delta\mathcal{E}^{(k)} = \mathcal{E}^{(k+1)} - \mathcal{E}^{(k)}. \tag{F.29}$$

Rewriting $\mathcal{B}^{(k+1)} \times_3 D^{(k)} + \mathcal{E}^{(k)} - \mathcal{Y}$ using the updated variables $(k+1)$:

$$\begin{aligned}&\mathcal{B}^{(k+1)} \times_3 D^{(k)} + \mathcal{E}^{(k)} - \mathcal{Y} \\ &\quad = \left[\mathcal{B}^{(k+1)} \times_3 D^{(k+1)} + \mathcal{E}^{(k+1)} - \mathcal{Y}\right] \\ &\qquad + \left[\mathcal{B}^{(k+1)} \times_3 (D^{(k)} - D^{(k+1)}) + (\mathcal{E}^{(k)} - \mathcal{E}^{(k+1)})\right] \\ &\quad = \left[\mathcal{B}^{(k+1)} \times_3 D^{(k+1)} + \mathcal{E}^{(k+1)} - \mathcal{Y}\right] - \left[\mathcal{B}^{(k+1)} \times_3 \Delta D^{(k)} + \Delta\mathcal{E}^{(k)}\right].\end{aligned} \tag{F.30}$$

Step 5: Applying the Lagrangian multiplier update rule Based on the multiplier update rule (F.5):

$$\Lambda^{(k+1)} = \Lambda^{(k)} + \mu^{(k)} \left(\mathcal{Y} - \mathcal{B}^{(k+1)} \times_3 D^{(k+1)} - \mathcal{E}^{(k+1)}\right). \tag{F.31}$$

This implies:

$$\mu^{(k)} \left[\mathcal{B}^{(k+1)} \times_3 D^{(k+1)} + \mathcal{E}^{(k+1)} - \mathcal{Y}\right] = \Lambda^{(k)} - \Lambda^{(k+1)}. \tag{F.32}$$

Substituting this back into Eq. (F.28) and letting $\xi^{(k)} \in \partial\mathrm{Sep}(\mathcal{B}^{(k+1)}, \Omega, \vartheta)$:

$$\begin{aligned}\Lambda^{(k)} \times_3 (D^{(k)})^T = \xi^{(k)} + (\Lambda^{(k)} - \Lambda^{(k+1)}) \times_3 (D^{(k)})^T \\ - \mu^{(k)} \left[\mathcal{B}^{(k+1)} \times_3 \Delta D^{(k)} + \Delta\mathcal{E}^{(k)}\right] \times_3 (D^{(k)})^T.\end{aligned} \tag{F.33}$$

Simplifying ($\Lambda^{(k)}$ terms cancel):

$$\Lambda^{(k+1)} \times_3 (D^{(k)})^T = \xi^{(k)} - \mu^{(k)} \left[\mathcal{B}^{(k+1)} \times_3 \Delta D^{(k)} + \Delta\mathcal{E}^{(k)}\right] \times_3 (D^{(k)})^T. \tag{F.34}$$

Step 6: Adjusting for the transform matrix update Add and subtract $\Lambda^{(k+1)} \times_3 (D^{(k+1)})^T$ on the left:

$$\begin{aligned}\Lambda^{(k+1)} \times_3 (D^{(k+1)})^T - \Lambda^{(k+1)} \times_3 (\Delta D^{(k)})^T \\ = \xi^{(k)} - \mu^{(k)} \left[\mathcal{B}^{(k+1)} \times_3 \Delta D^{(k)} + \Delta\mathcal{E}^{(k)}\right] \times_3 (D^{(k)})^T.\end{aligned} \tag{F.35}$$

Rearranging yields the exact KKT form:

$$\Lambda^{(k+1)} \times_3 (D^{(k+1)})^T = \xi^{(k)} + \Psi^{(k)}, \tag{F.36}$$

where $\Psi^{(k)}$ denotes the total dual residual:

$$\Psi^{(k)} := \Lambda^{(k+1)} \times_3 (\Delta D^{(k)})^T - \mu^{(k)} \left[\mathcal{B}^{(k+1)} \times_3 \Delta D^{(k)} + \Delta\mathcal{E}^{(k)}\right] \times_3 (D^{(k)})^T. \tag{F.37}$$

Step 7: Taking the limit along the convergent subsequence Consider the convergent subsequence $\{k_j\}_{j=1}^{\infty}$ such that $(\mathcal{B}^{(k_j)}, D^{(k_j)}, \mathcal{E}^{(k_j)}, \Lambda^{(k_j)}) \to (\mathcal{B}^*, D^*, \mathcal{E}^*, \Lambda^*)$. As the algorithm converges, the differences between variables asymptotically vanish:

$$\|\Delta D^{(k_j)}\|_F \to 0, \quad \|\Delta\mathcal{E}^{(k_j)}\|_F \to 0 \quad \text{as } j \to \infty. \tag{F.38}$$

Assuming the sequences are bounded and noting that the residual $\Psi^{(k_j)}$ consists of terms multiplied by $\Delta D^{(k)}$ and $\Delta\mathcal{E}^{(k)}$:

$$\|\Psi^{(k_j)}\|_F \to 0 \quad \text{as } j \to \infty. \tag{F.39}$$

Taking the limit as $j \to \infty$ in Eq. (F.36):

- Left-hand side: $\Lambda^{(k_j+1)} \times_3 (D^{(k_j+1)})^T \to \Lambda^* \times_3 (D^*)^T$.
- Residual: $\Psi^{(k_j)} \to 0$.
- Subgradient: By the outer semi-continuity of the limiting subdifferential operator $\partial\text{Sep}(\cdot)$, the limit of $\xi^{(k_j)}$ belongs to $\partial\text{Sep}(\mathcal{B}^*, \Omega, \vartheta)$.

Therefore, we obtain the stationarity condition for $\mathcal{B}$:

$$0 \in \partial\text{Sep}(\mathcal{B}^*, \Omega, \vartheta) - \Lambda^* \times_3 (D^*)^T. \tag{F.40}$$

Step 8: Proving the continuity of Sep: Based on the structural expression of the convex-concave separation function and the Lipschitz continuity of singular value functions:

$$\begin{aligned}\left|\text{Sep}(\mathcal{B}^{(k+1)}, \Omega, \vartheta) - \text{Sep}(\mathcal{B}^*, \Omega, \vartheta)\right| &\leq \sum_{i=1}^{r_3} \vartheta_i \left|\text{Sep}(\tilde{\mathcal{B}}_{(i)}^{(k+1)}, \Omega) - \text{Sep}(\tilde{\mathcal{B}}_{(i)}^*, \Omega)\right| \\ &\leq C_{\text{Sep}} \|\mathcal{B}^{(k+1)} - \mathcal{B}^*\|_F,\end{aligned} \tag{F.41}$$

where $C_{\text{Sep}} = \sqrt{r_3}\|\vartheta\|_\infty(\omega + \|\tilde{\omega}\|_1)$. Since $\mathcal{B}^{(k_j)} \to \mathcal{B}^*$, we have $\text{Sep}(\mathcal{B}^{(k_j)}, \Omega, \vartheta) \to \text{Sep}(\mathcal{B}^*, \Omega, \vartheta)$. By the outer semi-continuity of the limiting subdifferential (Lemma 15), as $k_j \to \infty$:

$$0 \in \partial \text{Sep}(\mathcal{B}^*, \Omega, \vartheta) - \Lambda^* \times_3 (D^*)^T. \tag{F.42}$$

(III) Stationarity with respect to $\mathcal{E}$

From Fermat's Rule for the $\mathcal{E}$-subproblem (F.4):

$$0 \in \partial \left(\lambda \|\mathcal{E}^{(k+1)}\|_1 + \frac{\mu^{(k)}}{2} \left\| \mathcal{E}^{(k+1)} - \left(\mathcal{Y} - \mathcal{B}^{(k+1)} \times_3 D^{(k+1)} + \frac{\Lambda^{(k)}}{\mu^{(k)}} \right) \right\|_F^2 \right). \tag{F.43}$$

By the sum rule:

$$0 \in \lambda \partial \|\mathcal{E}^{(k+1)}\|_1 + \mu^{(k)}(\mathcal{E}^{(k+1)} - \mathcal{Y} + \mathcal{B}^{(k+1)} \times_3 D^{(k+1)} - \Lambda^{(k)}/\mu^{(k)}). \tag{F.44}$$

Rearranging yields:

$$\Lambda^{(k)} + \mu^{(k)}(\mathcal{Y} - \mathcal{B}^{(k+1)} \times_3 D^{(k+1)} - \mathcal{E}^{(k+1)}) \in \lambda \partial \|\mathcal{E}^{(k+1)}\|_1. \tag{F.45}$$

By the multiplier update formula (F.5):

$$\Lambda^{(k+1)} \in \lambda \partial \|\mathcal{E}^{(k+1)}\|_1. \tag{F.46}$$

By outer semi-continuity, as $k_j \to \infty$:

$$\Lambda^* \in \lambda \partial \|\mathcal{E}^*\|_1 \implies 0 \in \lambda \partial \|\mathcal{E}^*\|_1 - \Lambda^*. \tag{F.47}$$

(IV) Primal Feasibility and Constraints

From Theorem 3:

$$\lim_{k \to \infty} \|\mathcal{Y} - \mathcal{B}^{(k+1)} \times_3 D^{(k+1)} - \mathcal{E}^{(k+1)}\|_F = 0. \tag{F.48}$$

Taking the limit gives:

$$\mathcal{Y} = \mathcal{B}^* \times_3 D^* + \mathcal{E}^*. \tag{F.49}$$

Since $(D^{(k)})^T D^{(k)} = I_{r_3}$ holds for all k and the column orthogonal constraint set is closed:

$$(D^*)^T D^* = I_{r_3}. \tag{F.50}$$

(V) Stationarity with respect to D

Step 1: Optimality condition of the D-subproblem The D-subproblem (F.3) is equivalent to the Procrustes problem:

$$D^{(k+1)} = \arg \max_{D^T D = I_{r_3}} \operatorname{tr}(D^T M^{(k)}), \tag{F.51}$$

where:

$$M^{(k)} = [\operatorname{unfold}_3 \left(\mathcal{Y} - \mathcal{E}^{(k)} + \Lambda^{(k)}/\mu^{(k)}\right)]^T \cdot \operatorname{unfold}_3(\mathcal{B}^{(k+1)}). \tag{F.52}$$

Step 2: KKT conditions on the Stiefel manifold D is an optimal solution for the Procrustes problem if and only if there exists a symmetric matrix Λ_D such that:

$$M = D\Lambda_D, \tag{F.53}$$

where $\Lambda_D = D^T M$ is the Lagrangian multiplier. From the closed-form solution, $D^{(k+1)} = \operatorname{Polar}(M^{(k)})$, satisfying:

$$M^{(k)} = D^{(k+1)} \Lambda_D^{(k)}, \tag{F.54}$$

where $\Lambda_D^{(k)} = (D^{(k+1)})^T M^{(k)}$ is a symmetric positive definite matrix.

Step 3: Taking the limit along the convergent subsequence For the subsequence $\{k_j\}$:

$$(\mathcal{B}^{(k_j)}, D^{(k_j)}, \mathcal{E}^{(k_j)}, \Lambda^{(k_j)}) \to (\mathcal{B}^*, D^*, \mathcal{E}^*, \Lambda^*). \tag{F.55}$$

The limit of $M^{(k_j)}$ is:

$$M^{(k_j)} \to M^* = \operatorname{unfold}_3 \left(\mathcal{Y} - \mathcal{E}^*\right)^T \cdot [\operatorname{unfold}_3(\mathcal{B}^*)], \tag{F.56}$$

since $\mathcal{E}^{(k_j)} \to \mathcal{E}^*$, $\Lambda^{(k_j)}/\mu^{(k_j)} \to 0$ (due to the bounded Λ and $\mu^{(k_j)} \to \infty$), and $\mathcal{B}^{(k_j+1)} \to \mathcal{B}^*$.

Step 4: Continuity of polar decomposition Polar decomposition is continuous with respect to the matrix; if $M_n \to M$, then $\operatorname{Polar}(M_n) \to \operatorname{Polar}(M)$. Thus:

$$D^{(k_j+1)} = \operatorname{Polar}(M^{(k_j)}) \to \operatorname{Polar}(M^*) =: D^*. \tag{F.57}$$

Step 5: KKT conditions of the limit point
From $M^{(k_j)} = D^{(k_j+1)}\Lambda_D^{(k_j)}$, taking the limit gives:

$$M^* = D^*\Lambda_D^*, \tag{F.58}$$

where $\Lambda_D^* = (D^*)^T M^*$ is a symmetric matrix. This indicates that D^* satisfies the first-order optimality condition on the Stiefel manifold:

$$D^* = \text{Polar}(M^*), \quad M^* = \text{unfold}_3(\hat{\mathcal{Y}}^*)^T \cdot [\text{unfold}_3(\mathcal{B}^*)], \tag{F.59}$$

where $\hat{\mathcal{Y}}^* = \mathcal{Y} - \mathcal{E}^*$.

Step 6: Relation with primal feasibility From the primal feasibility in Theorem 3, $\mathcal{Y} = \mathcal{B}^* \times_3 D^* + \mathcal{E}^*$, so:

$$\text{unfold}_3(\mathcal{Y} - \mathcal{E}^*) = \text{unfold}_3(\mathcal{B}^*) \cdot (D^*)^T. \tag{F.60}$$

Substituting into the definition of M^* verifies:

$$M^* = \text{unfold}_3(\mathcal{B}^*) \cdot (D^*)^T \cdot [\text{unfold}_3(\mathcal{B}^*)]^T, \tag{F.61}$$

which is consistent with $D^* = \text{Polar}(M^*)$, confirming the completeness of the KKT conditions.

In summary, the limit point $(\mathcal{B}^*, D^*, \mathcal{E}^*, \Lambda^*)$ satisfies all generalized KKT conditions in (F.19). □

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